



Research paper

P4P: A probe-guided anti-poisoning defense for federated learning-based intrusion detection in IoT networks under non-IID data

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ABSTRACT

Federated learning (FL) enables collaborative Intrusion Detection Systems (IDS) across distributed Internet of Things (IoT) networks without sharing raw data. However, its openness exposes it to model poisoning and backdoor attacks, where malicious clients manipulate updates to corrupt the global model. Detecting such threats remains difficult under non-independent and identically distributed (non-IID) data, where benign updates diverge naturally and resemble adversarial behavior. Existing defenses based on robust aggregation, clustering, or cryptographic auditing either assume IID conditions, lack a global semantic reference, or incur high computational costs, leading to poor robustness and high false positives. To overcome these challenges, we propose probe-for-poisoning (P4P), a lightweight system-level integration and refinement of similarity-based defenses for securing FL aggregation. P4P constructs a probe vector from the global model's historical trajectory and actively queries clients. To establish trust, the server itself computes a single scalar probe response representing the cosine similarity between each client's received local gradient and the probe vector, ensuring all computationally intensive detection is performed centrally without relying on client-reported metrics. Combined with L2-norm filtering, ensemble anomaly detection, and temporal suspicion scoring, P4P forms a multi-metric defense pipeline capable of identifying both abrupt and stealthy poisoning behaviors. Experiments on IoTDIAD 2024 and ACI-IoT 2023 datasets show that P4P improves accuracy by up to 18% and reduces false positives by 25% compared to state-of-the-art defenses such as FedMP and Recess, with negligible overhead. These results highlight P4P as a robust and practical anti-poisoning solution for federated intrusion detection under realistic, resource-constrained IoT network scenarios.

1. Introduction

The rapid proliferation of interconnected sensing and control devices in modern Internet of Things (IoT) ecosystems, including smart environments, industrial monitoring nodes, and edge-deployed gateways, has dramatically increased the scale, diversity, and dynamics of network telemetry. As these distributed devices continuously exchange flow-level traffic and system signals, the attack surface expands correspondingly, posing substantial challenges for current intrusion detection systems (IDS) (Abdulganiyu et al., 2023) to address. Traditional, centrally trained IDS require aggregation of raw traffic to a central server, which creates privacy risks, regulatory obstacles, and heavy communication and storage costs-constraints that are especially problematic in bandwidth- and resource-limited edge/IoT deployments.

Federated learning (FL) (Li Li et al., 2020) mitigates these issues by enabling distributed clients to collaboratively train a shared detection model while keeping raw data local. Moreover, FL distributes the computational burden of processing complex heterogeneous data structures (e.g., dissimilarity metrics), which are often computationally infeasible to handle centrally (Belenguer et al., 2023). Thus, FL becomes an attractive paradigm for privacy-aware and scalable IDS deployments in the IoT context (Wang et al., 2024).

In the FL-based IDS setting, each participating IoT device or local gateway acts as a client that learns from its local traffic patterns and contributes model updates to a central aggregator (Bhavsar et al., 2024). This framework enables models to generalize across heterogeneous attacks while preserving data privacy. However, the open and decentralized nature of FL introduces new attack surfaces that

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traditional IDS do not face. Malicious participants can intentionally manipulate local gradients to launch model poisoning or backdoor attacks, degrading the detection performance (availability attacks) or implanting hidden triggers to evade detection (integrity attacks) (Bhagoji et al., 2019; Fang et al., 2021). Even a small fraction of compromised clients can destabilize convergence and cause persistent corruption of the global model (Shejwalkar and Houmansadr, 2021).

These vulnerabilities are further aggravated by the non-IID data distributions inherent to IoT environments, where traffic data collected by different devices vary significantly in size, label frequency, and feature representation (Zhang et al., 2025). In such heterogeneous settings, benign updates may naturally diverge from the global trend, often mimicking the statistical behavior of adversarial updates (Fang et al., 2021). Consequently, existing anomaly-based defenses such as clustering or outlier detection of client updates tend to misclassify benign clients as malicious, leading to high false positive rates and unstable learning dynamics.

Existing poisoning defenses in federated learning can be systematically categorized into four major groups, each exhibiting distinguished strengths but also suffering from fundamental limitations that hinder their effectiveness under realistic non-IID FL-IoT environments.

The first category, passive robust aggregation and cluster-aware filtering: This category adopts a statistical perspective and relies on detecting anomalous client updates by measuring deviation from the aggregation consensus. Traditional robust aggregation techniques such as, Trimmed Mean and Median (Yin et al., 2018) assume that malicious updates significantly differ from benign ones. More recent approaches, such as FedAMM (Gai et al., 2025), leverage gradient similarity metrics, while FedMP (Zhao et al., 2025) introduces cluster-aware aggregation to partially mitigate non-IID divergence by grouping semantically aligned clients. These methods incur no additional querying overhead and perform effectively in IID environments where benign updates are expected to remain statistically consistent. But, they fundamentally assume that benign clients remain mutually similar; however, in non-IID settings, benign updates may naturally diverge due to heterogeneous label distributions, varying sample sizes, and feature heterogeneity (Liu et al., 2023). The absence of a global semantic reference causes client-to-client comparisons to become unfair, leading to false flagging of honest clients. Although FedMP (Zhao et al., 2025) improves non-IID tolerance via clustering, it introduces substantial overhead and relies on static similarity metrics that remain vulnerable to adaptive or gradual poisoning strategies.

Shifting from passive observation to active detection, a second category involves proactive querying and reputation-based detection: These defenses shift from passive observation to active detection, probing clients through challenge-response mechanisms or tracking cumulative reliability via reputation models (Zhang et al., 2022). For example, RECESS (Yan et al., 2025) employs targeted active queries to test client consistency over time. Such methods improve resilience against stealthy or slow poisoning by dynamically assessing behavioral consistency rather than relying purely on statistical deviation (Liang et al., 2023). But, these techniques introduce additional communication and energy overhead (Carvalho et al., 2024), which may become prohibitive for resource-constrained IoT devices, IDS. Furthermore, reputation drift may unfairly penalize benign clients suffering from temporary instability due to non-IID (Liu et al., 2023). Adversaries may also adapt to probing patterns and align temporarily with expectations, evading long-term detection.

A third line of work, focusing on cryptographic and trust-anchored auditing mechanisms: secures update integrity via blockchain-assisted verification, secure multiparty computation, or zero-knowledge proofs. For instance, TMT-FL (Lu et al., 2025) and DPAD (Basak and Chatterjee, 2025) employ cryptographically enforced traceability and auditability to prevent tampering. These approaches offer strong security guarantees and protect against unauthorized manipulation, ensuring trustworthiness and accountability in update submission. While

they validate the structural correctness of submitted updates, they do not assess the semantic intention behind them, leaving integrity and backdoor poisoning undetected (Bagdasaryan et al., 2019). Moreover, cryptographic mechanisms introduce significant latency and computational burden, posing scalability challenges for latency-sensitive IoT-FL deployments (Bonawitz et al., 2016; Bao et al., 2019).

Finally, to improve resilience, hybrid multi-strategy and temporal monitoring frameworks: To improve resilience, hybrid strategies such as MSGuard (Yang et al., 2025), FLGuardian (Zhou et al., 2025), and FLANDERS (Gabrielli et al., 2025) combine multiple detection signals, including clustering, L2-filtering, anomaly scoring, and temporal monitoring. They improve detection robustness by incorporating multiple complementary heuristics and reduce reliance on a single decision criterion. Despite multi-stage validation, many hybrid systems remain computationally complex and operationally heavy. Moreover, most fail to model the temporal evolution of benign learning trajectories under global guidance. This limitation is consistent with recent surveys indicating that the transition from static batch learning to real-time streaming data analysis remains an underdeveloped area in current FL-IDS literature (Belenguer et al., 2025). This lack of temporal reasoning is a critical flaw, as the network traffic environment in FL-IDS is inherently dynamic and non-stationary. On one hand, benign user behavior and network patterns constantly evolve, a phenomenon known as concept drift. A temporally-blind defense cannot distinguish this natural evolution from anomalies, leading to a high rate of false positives. On the other hand, adversaries continuously adapt their strategies, a process known as attack pattern drift, to evade detection. Consequently, they may miss stealthy poisoning patterns (Xie et al., 2020) that gradually align with trends until triggering malicious objectives (Chen et al., 2024).

Limitations of existing defense paradigms. Although the aforementioned approaches have advanced the robustness of FL-based intrusion detection systems (IDS), they still exhibit several fundamental limitations rooted in their design principles. Specifically, robust aggregation and hybrid defenses suffer from (i) and (ii), while proactive and probing-based methods alleviate (iii) but remain constrained by (iv):

- (i) *Reliance on clean or globally consistent validation references under non-IID divergence.* Robust aggregation and clustering-based defenses (e.g., FedAMM (Gai et al., 2025), FedMP (Zhao et al., 2025)) inherently assume the availability of globally aligned validation data to distinguish benign from malicious updates. This assumption breaks down under heterogeneous data distributions, where benign deviations may be misclassified as poisoning.
- (ii) *Limited adaptability and computational inefficiency during active or trust-anchored adaptation.* Hybrid and cryptographic frameworks (e.g., MSGuard (Yang et al., 2025), FLGuardian (Zhou et al., 2025), TMT-FL (Lu et al., 2025), DPAD (Basak and Chatterjee, 2025)) employ complex aggregation and encryption pipelines that hinder timely adaptation to dynamic client behaviors, resulting in high computational overhead and limited scalability for real-time IDS deployment.
- (iii) *Insufficient semantic or contextual awareness of structural integrity.* Although proactive probing and reputation-based mechanisms (e.g., RECESS (Yan et al., 2025)) can capture behavioral deviations, they often overlook semantic poisoning, where malicious clients subtly alter feature-level representations without producing large gradient deviations.
- (iv) *Weak temporal reasoning and cross-round consistency.* Even advanced hybrid defenses (e.g., FLGuardian (Zhou et al., 2025), MSGuard (Yang et al., 2025)) typically rely on static evaluation strategies that lack temporal awareness to track client consistency across rounds, leading to degraded robustness against stealthy or temporally coordinated attacks.

Taken together, these limitations highlight the lack of a defense mechanism that simultaneously distinguishes malicious behavior from benign non-IID divergence while maintaining temporal robustness and remaining lightweight at the client side with minimal computation and communication overhead, and practical for deployment in resource-constrained IoT networks.

To bridge this gap, we propose P4P (Probe-for-Poisoning), a lightweight system-level integration and refinement of similarity-based defenses tailored for FL-based intrusion detection under non-IID environments. Unlike conventional passive approaches, P4P introduces an active probing mechanism that derives probe vectors from the temporal trajectory of the global model. These probes serve as a shared global reference, enabling the server to evaluate each client's update direction via cosine similarity with the global trajectory, thereby distinguishing benign heterogeneity from adversarial deviation. Crucially, the server recomputes this scalar similarity value directly from the received updates, while all computationally intensive detection and analysis are performed at the server side.

Furthermore, P4P integrates L2-norm filtering, ensemble anomaly detection, and a temporal suspicion scoring mechanism to track both abrupt and slow poisoning behaviors. This multi-layer design ensures temporal awareness and strong adaptability to non-IID data, while preserving low client-side overhead suitable for bandwidth- and energy-constrained IoT environments.

In summary, our main contributions are as follows:

- Proactive probe-based defense: We introduce a lightweight querying mechanism that evaluates client alignment with the global trajectory, enabling early detection of directional inconsistencies induced by poisoning.
- Temporal and non-IID robustness: Through probe chaining across rounds, P4P maintains temporal continuity and effectively mitigates confusion caused by non-IID distributions.
- Multi-layer lightweight architecture: The proposed defense integrates L2-norm filtering, ensemble anomaly detection, and temporal scoring to deliver robustness and scalability, while imposing minimal computation and communication overhead on IoT clients.
- Comprehensive evaluation: Experiments with real-world IoT intrusion detection datasets (IoTDIAD (Canadian Institute for Cybersecurity, 2024) and ACI-IoT (ACI and Collaborators, 2024)) demonstrate that P4P achieves superior defense performance against various poisoning attacks, including Sign-Flip, LIE, and Backdoor, while maintaining high accuracy and efficiency under heterogeneous non-IID conditions.

The remainder of this article is organized as follows. Section 2 presents the related work. Section 3 describes the proposed methodology in detail. Next, Section 4 presents the dataset and experimental setup, followed by Section 5 with performance evaluation and analysis. We discuss threats to validity in Section 6. Finally, Section 7 concludes the article and outlines future directions.

2. Related work

2.1. Federated Learning and Intrusion Detection Systems (IDS)

Federated Learning (FL) has been widely adopted in security-sensitive applications, particularly for building intrusion detection systems (IDS) in distributed IoT and edge environments. By allowing multiple clients to collaboratively train a global model without sharing raw data, FL preserves privacy while leveraging diverse datasets (Kairouz et al., 2021; Agrawal et al., 2022). However, the same decentralization that makes FL attractive also increases its vulnerability to adversarial manipulation. For IDS, where the correctness of anomaly detection is crucial, poisoned updates from a few compromised clients can

significantly degrade detection accuracy or even implant backdoors for stealthy intrusions (Liu et al., 2020). Recent advances, such as Fed-LSAE (Luong et al., 2024), highlight a shift toward latent-space inspection for FL-based IDS, leveraging feature representations instead of raw parameters to enhance robustness against internal poisoning.

2.2. Model poisoning attack

Among the threats to FL, model poisoning attacks are particularly dangerous. Unlike data poisoning, which corrupts training inputs, model poisoning directly manipulates gradients or weight updates. Attack vectors include sign-flipping, Gaussian perturbations, and label-flipping, as well as more sophisticated strategies such as Little is Enough (LIE), backdoors, and stealthy scaling attacks (Fang et al., 2021; Bhagoji et al., 2019; Baruch et al., 2019). These methods enable adversaries to either degrade global accuracy (untargeted Byzantine attacks) or inject malicious behavior into the global model (targeted backdoors). In Non-IID settings typical of IDS deployments, benign clients naturally produce divergent updates, making it even harder to distinguish between malicious and honest contributions (Shejwalkar and Houmansadr, 2021).

2.3. Secured aggregation and defense mechanisms

To counter poisoning, a broad range of defenses have been proposed. Passive approaches, such as FedMP (Zhao et al., 2025), FL-Guardian (Zhou et al., 2025), DPAD (Basak and Chatterjee, 2025), and MSGuard (Yang et al., 2025), rely on anomaly detection or clustering of submitted updates. While effective in controlled settings, they incur high computational overhead and often assume majority honesty, which is fragile under Non-IID distributions. Proactive defenses, such as RECESS (Yan et al., 2025), introduce active probes (e.g., gradient traps) but remain sensitive to adaptive attackers. Cryptographic methods, including TMT-FL (Lu et al., 2025), guarantee data integrity through zero-knowledge proofs or blockchain auditing, yet their costs are prohibitive and they do not directly detect semantic poisoning. More recent multi-layer defenses (e.g., FedAMM (Gai et al., 2025), FedCleanse (Huang et al., 2025), UDFed (Deng et al., 2025), FLANDERS (Gabrielli et al., 2025)) combine multiple signals such as reputation, gradient decomposition, or temporal anomaly detection. Similarly, pFL-IDS (Thein et al., 2024) employs a server-side detection mechanism that filters malicious updates based on their cosine similarity relative to a pre-computed centroid. In parallel, Fed-LSAE (Luong et al., 2024) introduces a lightweight latent-space inspection defense that filters clients via penultimate-layer representations, Autoencoder compression, and RBF-CKA clustering to exclude malicious participants, yet it assumes a minority of attackers and a fixed configuration. While these strategies improve robustness, they remain limited by scalability, high computational costs, or lack of proactive verification.

2.4. Non-IID issues

A fundamental limitation across most defenses is the Non-IID nature of federated datasets. IDS clients often monitor highly heterogeneous traffic patterns, leading to natural divergence of updates. Defenses that assume benign clients converge toward a majority direction (e.g., MSGuard (Yang et al., 2025), FL-Guardian (Zhou et al., 2025)) misclassify honest but divergent clients as adversaries, increasing false positives. Conversely, adaptive attackers can exploit this heterogeneity to camouflage poisoned updates (Shejwalkar and Houmansadr, 2021; Yang et al., 2025). Furthermore, most defenses evaluate updates independently at each round, without considering temporal consistency, leaving them vulnerable to stealthy or slow poisoning (Fang et al., 2021). Fed-LSAE (Luong et al., 2024) alleviates some Non-IID variance through Autoencoder-compressed latent representations but performs round-wise filtering without temporal adaptation.

2.5. Limitations of existing defense paradigms

Despite notable progress in federated learning defenses, current paradigms exhibit fundamental limitations that arise from their underlying design philosophies. These limitations manifest differently across the four main categories:

Robust Aggregation and Clustering-Based Methods: Approaches such as FedAMM (Gai et al., 2025), FedMP (Zhao et al., 2025), and pFL-IDS (Thein et al., 2024) rely on statistical consistency or geometric similarity among client updates. While recent works like pFL-IDS attempt to handle Non-IID heterogeneity through personalization and centroid-based filtering, they remain fundamentally passive defenses. By relying solely on observing the distribution of submitted updates (e.g., cosine similarity), these methods lack a proactive mechanism to verify client intent. Consequently, they cannot reliably distinguish between legitimate Non-IID deviation and stealthy adaptive attacks that manipulate their updates to statistically mimic the benign centroid.

Cryptographic and Audit-Based Mechanisms: Methods like TMT-FL (Lu et al., 2025) and DPAD (Basak and Chatterjee, 2025) ensure the structural integrity of updates by verifying that submitted gradients match committed values. However, they do not evaluate whether these updates align with the underlying optimization trajectory. Consequently, these methods remain blind to semantic poisoning embedded within statistically valid updates.

Proactive and Reputation-Based Defenses: Systems such as RECESS (Yan et al., 2025) address some issues by actively probing clients. However, such mechanisms introduce sensitivity to adaptive adversaries, who may manipulate behavior to temporarily mimic honest clients and evade detection. Furthermore, additional probing overhead can become prohibitive in resource-constrained IoT deployments. A subset of these defenses utilizes server-side trusted reference datasets to cross-validate client updates, a strategy notably applied in robust Android malware detection (Eslamnejad et al., 2025). While these methods are effective for identifying specific threats like label-flipping, they fundamentally rely on the availability of a clean, representative validation set to optimize sensitivity parameter of this mechanism. This assumption is often impractical in IoT-IDS scenarios, where data heterogeneity (non-IID) makes it difficult to maintain a universal reference set, and the central collection of even a small validation set increases privacy risks and computational burden.

Hybrid Frameworks: While combining multiple signals (e.g., MS-Guard (Yang et al., 2025), FLGuardian (Zhou et al., 2025)) improves robustness, these systems often suffer from high computational complexity and limited temporal awareness. As client behavior is typically assessed at the round level without modeling long-term consistency, stealthy adversaries can still bypass detection.

In light of these challenges, a new defense mechanism must (i) remain lightweight for deployment in edge-based IDS, (ii) operate without clean validation datasets, (iii) adapt to Non-IID environments, and (iv) incorporate temporal analysis to detect stealthy adversaries. Our proposed P4P framework directly addresses these requirements by leveraging probe vectors and temporal chaining in a proactive multi-layer design.

2.6. Pipeline overview

As summarized in Table 1, existing defenses either incur high computational costs, rely on clean validation data, or fail to handle Non-IID scenarios effectively. In contrast, our proposed P4P framework achieves low overhead, strong Non-IID robustness, and temporal resilience, making it a lightweight yet proactive multi-layer defense.

3. Methodology

3.1. Threat model and assumptions

We consider an FL setting for intrusion detection in networked IoT environments, where a central server coordinates multiple distributed clients (e.g., IoT gateways or edge nodes). The server is assumed to be honest-but-curious and is not compromised during training. A fraction of participating clients may be malicious and attempt to perform untargeted or targeted byzantine attacks by manipulating their local training data or model updates. Malicious clients may collude and possess full knowledge of the global model parameters and training protocol, including the probe generation mechanism. In addition to abrupt poisoning, we consider colluding adversaries that coordinate small, consistent perturbations over multiple rounds to gradually steer the global trajectory toward a malicious direction, exploiting the incremental nature of the probe vector. To counter such stealthy drift, our framework incorporates a temporal suspicion scoring mechanism, mentioned in Section 3.3.5, that accumulates anomaly signals over time, ensuring that even subtle, persistent deviations are eventually penalized. However, adversaries are assumed not to control the central server, tamper with the communication channels, or compromise secure aggregation primitives. This threat model reflects realistic deployment scenarios in practical federated IDS for IoT networks.

To make the threat model concrete, we explicitly identify three classes of probe-aware adaptive strategies that an informed adversary may employ, along with the specific layer of P4P that counters each:

1. **Projected backdoor attack:** An adaptive attacker with knowledge of v_i may craft a malicious update ΔW_m that satisfies a high cosine similarity constraint ($\cos(\Delta W_m, v_i) \geq \tau$) while injecting a backdoor trigger in an orthogonal subspace. However, deflecting gradient energy into orthogonal dimensions necessarily inflates the update's L2-norm, causing it to be flagged and rejected by the **Magnitude Filtering** stage before the directional probe is ever evaluated.
2. **Adaptive on-off attack:** An adversary may alternate between benign and malicious rounds to suppress its cumulative suspicion score below the expulsion threshold θ . The **Temporal Suspicion Scoring** mechanism is explicitly designed to counter this strategy: the score $s_i^t = \max(0, s_i^{t-1} + \delta_i^t - \gamma)$ decays slowly (governed by γ), ensuring that clients who repeatedly re-emerge with anomalous behavior accumulate a rising score that eventually exceeds θ .
3. **Gradual drift attack:** Colluding clients inject small, consistent perturbations each round to slowly align the probe vector v_{i+1} with a malicious direction. As demonstrated in the formal analysis in , the detection time for such a strategy is bounded by $T_{detect} \approx \theta / (P(\delta_i^t=1) - \gamma)$, ensuring eviction before the trajectory drifts significantly from the legitimate optimization manifold.

3.2. Overview of the proposed framework

The core objective of our P4P framework, is to provide a proactive, lightweight, and multi-layer defense against poisoning attacks in FL. Existing defense strategies either rely on static aggregation rules or complex cryptographic primitives. Static rules (e.g., Krum, Trimmed Mean) treat client updates independently at each round and are vulnerable to collusion or stealthy poisoning, while cryptographic approaches (e.g., secure multiparty computation or blockchain-based logging) incur prohibitive overhead and still do not directly address model update manipulation. To overcome these limitations, we design P4P with three guiding principles:

- **Proactivity:** Instead of passively observing updates, the server actively queries clients using probe vectors derived from the global trajectory.

Table 1
Comparison of federated learning defense mechanisms (simplified).

Method	Defense type	Attack type	Clean dData	Non-IID	Overhead	Temporal	Multi-layer
FedAMM (Gai et al., 2025)	Robust + Clustering	Backdoor (majority)	No	High	High	No	Yes
TMT-FL (Lu et al., 2025)	Cryptographic (ZKP)	Data/model poison	No	Low	High	Yes	Yes
RECESS (Yan et al., 2025)	Proactive + Robust	Model poison (MPA)	No	Medium	Medium	Yes	Part
DPAD (Basak and Chatterjee, 2025)	Audit-based	Data poisoning	Yes	Low	High	No	No
MSGuard (Yang et al., 2025)	Multi-strategy	Sign, ALIE, MinMax	No	Medium	High	No	No
FLGuardian (Zhou et al., 2025)	Layer detection	Backdoor (layer)	No	Medium	Medium	Yes	Yes
FedCleanse (Huang et al., 2025)	Post-agg, Neuron	Backdoor (replace)	No	High	Low	Yes	Yes
FedMP (Zhao et al., 2025)	Multi-pronged	Byzantine (SF,LIE)	No	High	Medium	No	No
FLANDERS (Gabrielli et al., 2025)	Pre-agg + Time-ser	Untargeted poison	No	High	High	Yes	Part
UDFed (Deng et al., 2025)	Hybrid Universal	Noise, LIE, Backdoor	No	Medium	Low	Yes	Yes
Fed-LSAE (Luong et al., 2024)	Latent-space (Pre-agg)	Label-flip, GAN, Weight scale	No	High	Low	Yes	Part
Android Eslamnejad et al. (2025)	Density	Label-Flip	No*	Medium	High	No	Part
P4P (ours)	Proactive + Multi	Universal (Byz+BD)	No	High	Low	Yes	Yes

*While theoretically data-agnostic, this method requires pre-labeled data to optimize sensitivity parameters like k , ϵ , and $MinPts$ Eslamnejad et al. (2025).

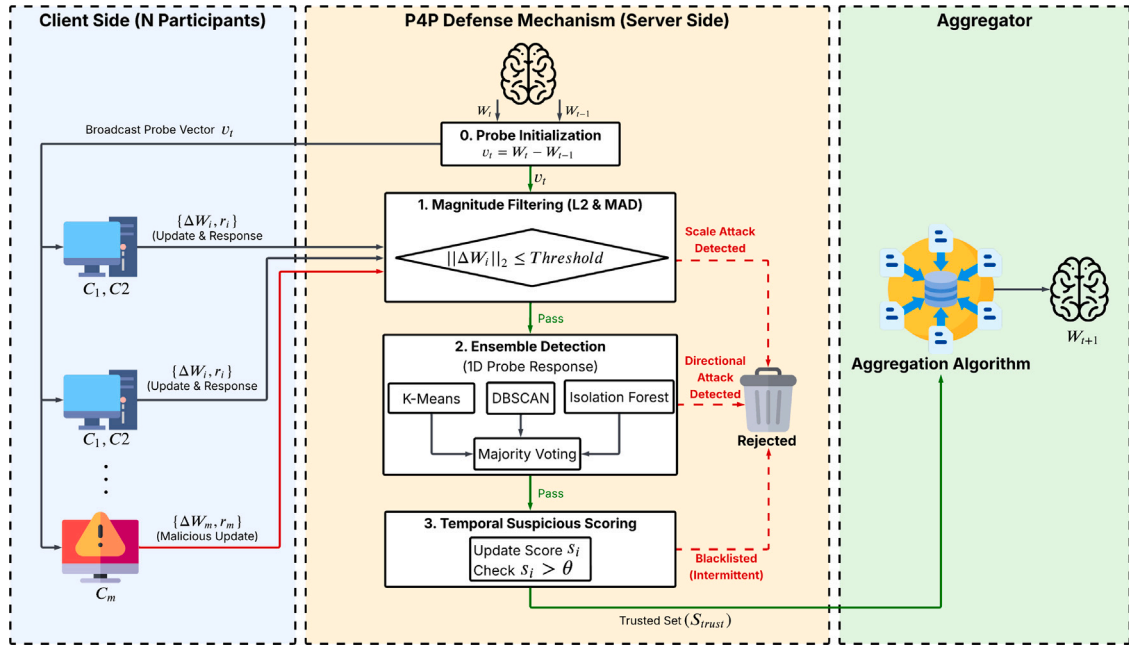


Fig. 1. Overview of the P4P framework workflow across one communication round.

- Temporal awareness: Malicious behaviors often manifest gradually or intermittently. By chaining probe vectors across rounds, the server leverages historical consistency to expose stealthy strategies.
- Lightweight robustness: Defense mechanisms must be practical in large-scale deployments. P4P transforms high-dimensional updates into low-dimensional probe responses, enabling efficient anomaly detection with minimal cost. Crucially, the server re-computes this scalar similarity value directly from the received updates, ensuring all computationally intensive detection and aggregation logic is executed centrally and mitigating the risk of clients fabricating benign-looking responses.

Fig. 1 illustrates the overall architecture of P4P. At each round, the server constructs a probe vector, receives client updates, centrally computes the probe responses, applies multi-layer detection, and finally aggregates only trusted contributions into the global model.

3.3. Proposed multi-layer defense mechanism

To ensure robustness against diverse adversarial strategies, P4P integrates a unified processing pipeline comprising four coordinated stages. First, Magnitude Filtering (Section 3.3.1) eliminates scale attacks by establishing dynamic thresholds based on statistical dispersion. Second,

the Probe-based Ensemble Detector (Sections 3.3.2–3.3.4) identifies directional poisoning by projecting updates onto a historical reference vector and applying orthogonal clustering algorithms. Third, Temporal Suspicion Scoring (Section 3.3.5) maintains a historical reputation for each client to mitigate intermittent or stealthy behaviors. Finally, the pipeline concludes with Robust Aggregation (Section 3.3.6), where only the validated updates are fused using a weighted protocol to evolve the global model.

3.3.1. L2-norm filtering

We first evaluate the scale of each update by its L2 norm:

$$\|\Delta W_i^t\|_2 = \sqrt{\sum_{j=1}^d (\Delta W_{i,j}^t)^2}. \quad (1)$$

Here, t denotes the current communication round, and i represents the client index. The term $\Delta W_{i,j}^t$ refers to the j th parameter element of the flattened update vector from client i , where j ranges from 1 to d with d representing the total dimensionality of the model parameters. We employ L2 norm filtering to defend against scale attacks, where adversaries artificially amplify the magnitude of their updates. Even if a malicious update aligns directionally with the global objective, an abnormally large magnitude allows it to disproportionately dominate the aggregation process. Without this filter, a single scaled-up update could

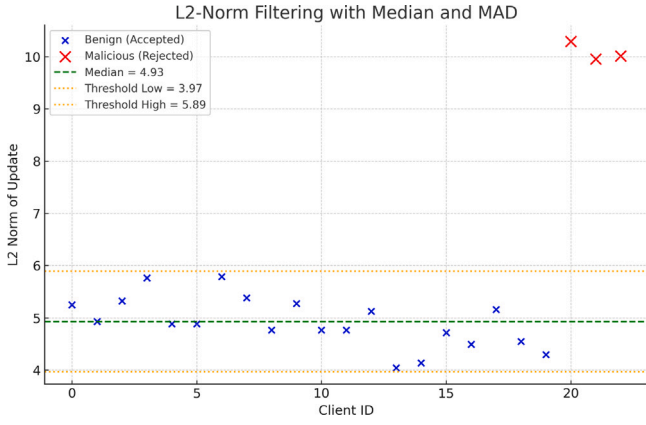


Fig. 2. Visualization of the Magnitude Filtering mechanism. The L2 norms of client updates are evaluated against a robust acceptance range defined by the Median Absolute Deviation (MAD).

overpower the contributions of legitimate clients, drastically shifting the global model weights towards the adversary's intended state. To robustly filter such outliers, we apply Median Absolute Deviation (MAD):

$$\text{MAD} = \text{median} \left(\left| \|\Delta W'_i\|_2 - \text{median}(\|\Delta W'\|_2) \right| \right). \quad (2)$$

We utilize MAD as a robust measure of statistical dispersion instead of the standard deviation. Since the mean and standard deviation are highly sensitive to extreme values, a single attacker with an arbitrarily large update could skew the threshold, causing the system to accept malicious inputs. In contrast, MAD relies on the median — a positional statistic — making the calculated threshold immune to the specific magnitude of extreme outliers, thereby preserving the filter's strictness. As shown in Fig. 2, the L2 norm of each client's model update is computed to measure its magnitude. To mitigate potential malicious contributions, the MAD statistic is used for outlier detection. As illustrated, clients marked in red are identified as outliers and rejected. Conversely, clients falling within the acceptance range—delimited by the median value (green dashed line) and the upper and lower thresholds derived from the MAD (orange dotted lines) are permitted to pass through this filtration layer. Updates exceeding a threshold (e.g., $3 \times \text{MAD}$) are temporarily flagged.

3.3.2. Probe vector construction

While L2-norm filtering effectively mitigates scale attacks by bounding the magnitude of updates, it remains blind to directional poisoning. Sophisticated adversaries can circumvent this filter by scaling their malicious updates to fall within the benign L2 range while still pointing in a deleterious direction (e.g., model steering or backdoor injection). To detect such angular deviations, we require a stable reference direction to evaluate the alignment of each client's update. This necessitates a stable reference direction to evaluate the alignment of each client's update. Addressing this critical requirement, the primary contribution of P4P lies in the choice of the *probe vector* and the surrounding decision pipeline, which functions as a dual-purpose anchor: it establishes a stable global reference to normalize client updates and enforces a temporal consistency chain derived from the model's historical trajectory.

Rather than relying on the vulnerable assumptions that the majority of clients are benign or that client behavior remains static—an oversight that leaves systems exposed to stealthy 'on-off' attacks where adversaries alternate between benign and malicious contributions—the server instead relies on its own inertia as a trust anchor. This effectively flags sudden behavioral shifts, even if the attacker was previously compliant.

Given the global parameters $W_t \in \mathbb{R}^d$ at round t , the probe vector is defined as:

$$v_t = W_t - W_{t-1}. \quad (3)$$

This formulation captures the *incremental direction* of model evolution between two consecutive rounds. It has two desirable properties: (1) it *encodes temporal history*, meaning that adversaries cannot easily alter it without consistently corrupting many rounds; (2) it is *model-dependent but data-agnostic*, requiring no access to raw data or trusted clean datasets.

Existing defense methodologies typically converge into two distinct paradigms, both with theoretical and practical constraints. First, data-dependent strategies leveraging server-side validation or trusted reference datasets are often infeasible in decentralized environments (clean data is unavailable) or exhibit representative bias (data does not capture client heterogeneity). Second, decentralized statistical frameworks (e.g., calculating pairwise similarities between clients) typically operate in a temporal-agnostic manner, analyzing updates solely within the current communication round. These methodologies demonstrate limited efficacy in identifying gradual drift attacks where adversaries induce a progressive optimization drift over time, as the malicious updates remain internally consistent in the current round but deviate from the historical trajectory.

By contrast, the probe vector leverages the server's own temporal inertia as a reference, requiring neither external data nor trust in the current round's majority. If malicious clients attempt to collude and manipulate the global trajectory to align with their malicious direction, they must first successfully inject their updates into W_{t+1} to shift the future probe vector ($v_{t+1} = W_{t+1} - W_t$). Because the subsequent anomaly detectors heavily penalize directional outliers relative to the current probe v_t , malicious clients are forced to align closely with the benign trajectory to avoid immediate rejection. Consequently, if their deviation is small enough to bypass the spatial filters, their influence on shifting the next probe is mathematically bounded and minimal.

Formal analysis of probe reliability. The core premise of P4P is grounded in the principle of Global Descent Consensus. Under non-IID conditions, while local gradients may diverge in magnitude and feature weights, benign clients still collaboratively minimize a shared global loss function. Consequently, their updates reside within a descent cone centered around the negative global gradient, maintaining a positive correlation with the global optimization direction.

This reliability is predicated on two core assumptions: (i) Gradual Trajectory, where $W_t - W_{t-1}$ provides a valid first-order approximation of the descent direction; and (ii) Directional Honesty, assuming the aggregate momentum of accepted updates from previous rounds points toward the task-specific manifold.

Under common FL assumptions — bounded gradient variance σ^2 , bounded client heterogeneity G , learning rate η , and T communication rounds — the expected deviation of the probe vector from the ideal case can be bounded as follows:

$$\mathbb{E}[\|v_t - v_t^*\|_2] \leq f \cdot \eta \cdot G + O\left(\frac{\sigma}{\sqrt{T}}\right), \quad (4)$$

where G captures the bounded gradient norm or heterogeneity term (as in standard FedAvg convergence analyses (McMahan et al., 2017)). The term $f \cdot \eta \cdot G$ reflects the maximum influence of malicious updates on the global trajectory after filtering, while the $O(1/\sqrt{T})$ term arises from stochastic gradient descent variance reduction over rounds.

For $f < 0.5$, the deviation $\epsilon = f \cdot \eta \cdot G + O(1/\sqrt{T})$ remains small relative to the typical scale of benign updates, ensuring that v_t continues to serve as a stable directional reference for cosine similarity evaluation. This bound holds even when malicious clients collude, because the multi-layer filtering mechanism (particularly temporal suspicion scoring) progressively excludes clients exhibiting persistent directional inconsistencies, limiting their cumulative impact on W_t .

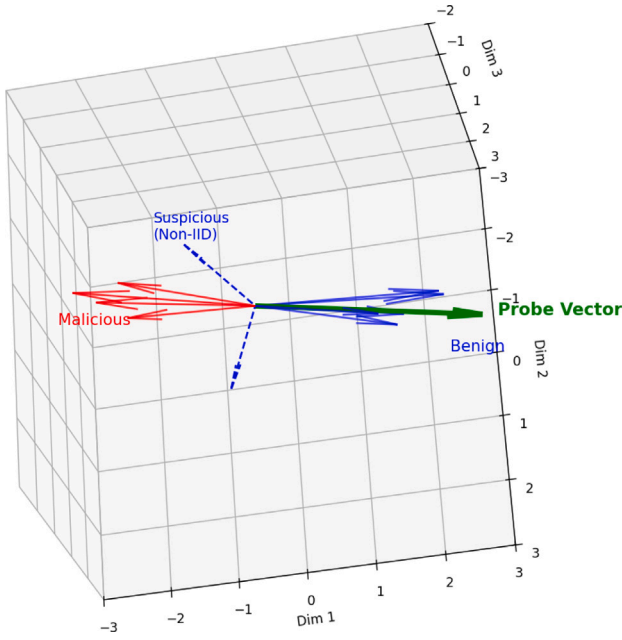


Fig. 3. Geometric intuition of the probe-based defense and clustering-based separation.

Regarding gradual steering attacks, where colluding adversaries inject small, coordinated perturbations over multiple rounds to slowly align the probe vector with a malicious direction, the temporal suspicion scoring mechanism provides effective mitigation. As defined in Eq. (6), the suspicion score is updated as $s_i^t = \max(0, s_i^{t-1} + \delta_i^t - \gamma)$, where $\delta_i^t \in \{0, 1\}$ is the binary anomaly indicator from the ensemble detector and γ is the decay factor.

To stealthily steer the model, an attacker must inject a small perturbation magnitude $\epsilon \ll 1$ at each round. However, persistently pushing the boundary of the benign distribution increases their likelihood of being flagged by the spatial filters (K-Means/DBSCAN). Assuming the probability of triggering a flag is proportional to the perturbation magnitude ($P(\delta_i^t = 1) \propto \epsilon$), the expected score accumulation per round is $\mathbb{E}[\Delta s_i^t] = c \cdot \epsilon - \gamma$ (where c is a proportionality constant). For an attack to successfully cause an accumulated drift, it requires $c \cdot \epsilon > \gamma$. Under this condition, the accumulated score grows linearly, and detection occurs within $t \approx \frac{\theta}{c \cdot \epsilon - \gamma}$ rounds once s_i^t exceeds the threshold θ . Consequently, the detection delay scales as $O(1/\epsilon)$, forcing adversaries into a strict trade-off: larger perturbations (ϵ) lead to rapid exclusion, while infinitesimally small perturbations fail to overcome the decay factor γ , rendering the attack mathematically bounded and ineffective.

Empirical evidence supporting this analysis is presented in Section 5.7, where we evaluate P4P under gradual collusion attacks with $f = 0.3$ – 0.4 , showing accuracy degradation of less than 5% compared to abrupt poisoning scenarios.

In summary, the probe vector remains a reliable trust anchor for $f < 0.5$, leveraging the inertia of the server-side trajectory and the multi-layer defense design, without requiring access to clean validation datasets or imposing additional client-side overhead.

3.3.3. Server-side probe response computation

To leverage the stable global reference established by the probe vector, we must first project each client's high-dimensional update onto this historical optimization trajectory. Instead of relying on client-reported metrics, which could be trivially fabricated by malicious actors to appear benign, the server computes a probe response r_i^t for each client C_i upon receiving its local update ΔW_i^t . This is done by

measuring the cosine similarity between the received update and the server-constructed probe vector v_t :

$$r_i^t = \frac{\langle \Delta W_i^t, v_t \rangle}{\|\Delta W_i^t\|_2 \cdot \|v_t\|_2}. \quad (5)$$

This transformation offers two critical advantages for the defense mechanism:

- **Scalability:** It compresses a model with millions of parameters into a one-dimensional similarity measure per client. This drastic dimensionality reduction enables the subsequent anomaly detection algorithms to operate with negligible computational overhead.
- **Normalization:** Since all clients are evaluated against the same reference v_t , the metric is normalized relative to the global momentum, mitigating the scale discrepancies often found in raw gradients.

From an optimization perspective, r_i^t quantifies the alignment with the global descent direction. However, interpreting this metric is challenging under Non-IID conditions. Ideally, benign updates should align perfectly ($r_i^t \approx 1$). In reality, due to local data heterogeneity, benign clients often exhibit high variance, resulting in lower or fluctuating similarity scores. Meanwhile, sophisticated adversaries may craft attacks with moderate similarity to blend in with this Non-IID noise. Therefore, a simple thresholding approach (e.g., rejecting all negative values) is insufficient and prone to high false positives. Instead, we must analyze the relative distribution of these scores to distinguish natural Non-IID variance from adversarial patterns, as detailed in the next section.

3.3.4. Probe-based anomaly detection

To ensure the global model converges towards the true objective, our ensemble focuses on the directional alignment of client updates using the client's probe response. This metric is crucial for detecting model poisoning attacks where adversaries attempt to steer the model's optimization trajectory without exhibiting magnitude anomalies.

As visualized in Fig. 3, the probe response creates a clear geometric separation between client groups. Benign clients align with the probe, forming a dense cluster with high similarity. In contrast, suspicious clients deviate due to affect of Non-IID environment, and malicious clients cluster in the opposite direction or behave as outliers. Motivated by this separability, we employ an ensemble of three detectors:

- **K-Means clustering:** acts as a consensus filter, separating clients into groups based on their angular deviation from the global model. It effectively segregates the majority — benign clients sharing a common descent direction — from divergent adversarial groups.
- **DBSCAN:** enforces density consistency. Since malicious updates often exhibit high variance in their directional impact or struggle to form a tight cluster compared to benign updates, DBSCAN identifies and removes these low-density directional outliers as noise.
- **Isolation Forest:** provides a statistical safeguard by analyzing the distribution of cosine scores. It isolates updates with rare alignment patterns that might evade cluster-based detection, ensuring that only clients with statistically typical directional contributions are aggregated.

As detailed in Algorithm 1, the ensemble voting mechanism consists of three main steps. First, K-Means focuses on global position, efficiently identifying the majority consensus but potentially failing against tightly clustered Sybil attacks. In the next step, we utilize DBSCAN to address this by enforcing local density constraints, rejecting attackers that mimic the global position but lack the cohesive structure of benign updates. However, both clustering methods rely on spatial proximity.

Algorithm 1 Ensemble Anomaly Detection

Require: Probe responses $\{r_i^t\}_{i \in S_{mag}}$, Client IDs $\{C_i\}$
Ensure: Suspicious client set S_{anom}

Initialize: $votes[i] \leftarrow 0$ for all $i \in S_{mag}$

K-Means Clustering $\triangleright$ Detecting directional outliers
Cluster $\{r_i^t\}$ into two clusters and obtain labels L^{kmeans}
Determine minority cluster $\mathcal{M}$
for $i = 1$ **to** $|S_{mag}|$ **do**
 if $L_i^{kmeans} \in \mathcal{M}$ **then**
 $votes[i] \leftarrow votes[i] + 1$
 end if
end for

DBSCAN $\triangleright$ Density-based flow analysis
Run DBSCAN on $\{r_i^t\}$ and obtain labels L^{dbscan}
for $i = 1$ **to** $|S_{mag}|$ **do**
 if $L_i^{dbscan} = -1$ **then**
 $votes[i] \leftarrow votes[i] + 1$
 end if
end for

Isolation Forest $\triangleright$ Tree-based isolation of malicious updates
Fit Isolation Forest on $\{r_i^t\}$ and obtain labels L^{iso}
for $i = 1$ **to** $|S_{mag}|$ **do**
 if $L_i^{iso} = 1$ **then** $\triangleright$ Client predicted as outlier
 $votes[i] \leftarrow votes[i] + 1$
 end if
end for

$S_{anom} \leftarrow \{i \mid votes[i] \geq 2\}$ $\triangleright$ Majority voting for robustness
return S_{anom}

Isolation Forest complements them by detecting statistical anomalies based on distribution paths, capturing stealthy attacks that might be spatially close to the center but exhibit rare feature patterns. Each detector provides a vote, and a client is considered suspicious if flagged by at least two. This ensemble approach increases robustness against adaptive attacks designed to bypass a single detector.

3.3.5. Temporal suspicion scoring

While the probe-based ensemble detector effectively identifies anomalies within a single round, it operates in a memoryless manner. Sophisticated adversaries may employ intermittent poisoning strategies behaving benignly in most rounds to lower their profile and attacking only sporadically to evade detection. To counter such time-varying threats, including attempts by colluding clients to incrementally drift the model trajectory over multiple rounds to avoid single-round detection, we introduce a temporal suspicion scoring mechanism that maintains a historical reputation for each client to address intermittent poisoning and reduce false positives, we introduce a suspicion score s_i^t :

$$s_i^t = \max(0, s_i^{t-1} + \delta_i^t - \gamma), \quad (6)$$

where $\delta_i^t = 1$ if client C_i is flagged at round t and 0 otherwise, and γ is a decay factor allowing benign clients to recover over time. If $s_i^t > \theta$, the client is permanently removed.

Adversarial drift Analysis: To evaluate the robustness against gradual steering, consider a set of f colluding malicious clients injecting a small perturbation δ towards a malicious direction $\vec{m}$ in each round. The global update becomes

$$\Delta W_t \approx (1 - f)\Delta W_{benign} + f \cdot \delta \vec{m} \quad (7)$$

While the adversaries may choose δ small enough such that the anomaly score a_i^t remains slightly below the instantaneous detection threshold, the consistent direction of $\delta \vec{m}$ over T rounds creates a cumulative drift in the probe vector $v_t \approx \sum_{k=1}^t \Delta W_k$. However, because

the ensemble detector incorporates multiple metrics, a persistent directional bias will statistically increase the probability of $\delta_i^t = 1$ over time. The detection time T_{detect} for a steering attack is bounded by:

$$T_{detect} \approx \frac{\theta}{P(\delta_i^t = 1) - \gamma} \quad (8)$$

where $P(\delta_i^t = 1)$ is the probability of being flagged. As long as the attack is persistent enough to steer the model, $P(\delta_i^t = 1) > \gamma$, ensuring the adversary is eventually evicted before the global trajectory aligns significantly with $\vec{m}$.

This mechanism leverages temporal history, ensuring that malicious behavior accumulating over rounds is detected, while benign clients with occasional deviations are not unfairly penalized. Even if colluding adversaries occasionally bypass the spatial filters with minor directional deviations, their accumulated attempts to steer the model will inevitably trigger this temporal tripwire, resulting in their permanent exclusion.

3.3.6. Robust aggregation

Finally, after filtering magnitude outliers, voting on directional anomalies, and checking historical reputation, the server proceeds to aggregate the updates. To ensure that only benign contributions shape the future model, we adopt a weighted Federated Averaging (FedAvg) (McMahan et al., 2017) scheme:

$$W_{t+1} = W_t + \eta \cdot \frac{\sum_{i=1}^N \omega_i^t \Delta W_i^t}{\sum_{i=1}^N \omega_i^t}, \quad (9)$$

where η represents the global learning rate, and $\omega_i^t \in \{0, 1\}$ is a trust indicator derived from the preceding defense layers (i.e., $\omega_i^t = 1$ if client C_i passes all filters, and 0 otherwise). Generality of the Protocol: It is important to highlight that P4P is designed as a modular security layer, decoupled from the specific aggregation rule. Since the defense mechanism operates solely on the geometry and statistics of the update vectors to identify the trusted set S_{trust} , it is mathematically compatible with any aggregation scheme. While we utilize FedAvg (Eq. (9)) in our evaluation as the representative baseline, the filtered set can be seamlessly integrated with other aggregation algorithms, such as FedSGD (Wei et al., 2021), FedProx (Tian Li et al., 2020), or FedNova (Wang et al., 2020), without modifying the core defense logic. By integrating these multiple layers of defense before the aggregation step, P4P effectively minimizes the influence of malicious updates, maintaining the integrity of the global model even under sophisticated adaptive attacks.

3.3.7. Convergence analysis

To address the impact of partial filtering on the global model, we analyze the convergence properties of P4P under non-IID conditions. FedAVG in heterogeneous environments typically follows a convergence rate of $O(1/T)$ (McMahan et al., 2017) with an additional term representing data heterogeneity. P4P preserves these guarantees through the following mechanisms:

- **Trust Anchor Consistency:** Unlike passive defenses that rely on peer-to-peer comparisons, P4P utilizes the server's historical trajectory ($v_t = W_t - W_{t-1}$) as a stable reference. By ensuring that selected updates (S_{trust}) align with the global descent direction, P4P effectively reduces gradient variance caused by malicious outliers without shifting the optimization target of benign clients.
- **Bounded Filtering Bias:** Theoretical convergence remains intact if the filtering mechanism identifies malicious updates while retaining a representative majority of benign contributions. Since P4P's "No Attack" performance at extreme non-IID levels ($\alpha = 0.1$) remains within 2.68% of the undefended baseline (87.55% vs 90.23%), the empirical bias introduced by our multi-layer filtering is mathematically negligible.

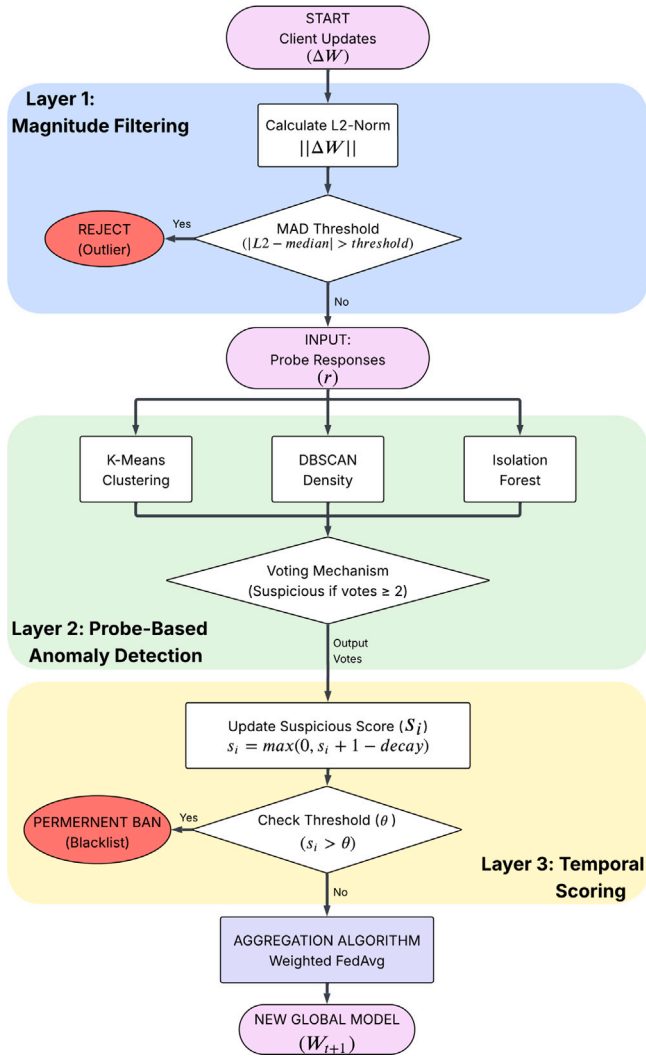


Fig. 4. The workflow of the multi-stage defense framework utilizing L2-norm filtering, ensemble clustering, and reputation scoring to mitigate poisoning attacks.

- **Error Bound Stability:** Under the assumption that the malicious fraction $f < 50\%$, the expected global loss satisfies the bound

$$E[F(W_T) - F^*] \leq O(1/T) + \Gamma \quad (10)$$

where Γ is the non-IID drift. P4P's temporal scoring further stabilizes this bound by ensuring that intermittent poisoning does not accumulate enough momentum to derail the convergence trajectory.

Importantly, P4P does not modify the optimization objective of FedAvg. The filtering mechanism only removes suspicious updates while preserving the majority of benign gradients, therefore the convergence order remains identical to FedAvg under non-IID conditions.

3.4. Workflow summary

The P4P framework operates as a proactive, multi-layer security pipeline designed to sanitize client updates before they impact the global model, as illustrated in Fig. 4 and formally detailed in Algorithm 2. The workflow proceeds sequentially through five key phases:

1. **Magnitude Filtering (L2-norm Analysis):** Upon receiving local gradients, the system computes the L2 norm of each update and

Algorithm 2 The P4P Defense Mechanism

Require: Global model W_t, W_{t-1} , Client updates $\{\Delta W_i^t\}_{i=1}^N$, Client IDs $\{C_i\}_{i=1}^N$, Historical scores $\{s_i^{t-1}\}_{i=1}^N$
Ensure: Updated global model W_{t+1} , Updated scores $\{s_i^t\}_{i=1}^N$

Server Initialization:

$v_t \leftarrow W_t - W_{t-1}$ ▷ Probe Construction

$S_{mag} \leftarrow \emptyset, S_{trust} \leftarrow \emptyset$

Compute norms $L = \{\|\Delta W_i^t\|_2\}_{i=1}^N$

Calculate $T_{low}, T_{high} \leftarrow \text{MAD}(L)$

Phase 1: Magnitude Filtering

for $i = 1$ to N do

if $T_{low} \leq \|\Delta W_i^t\|_2 \leq T_{high}$ then

$S_{mag} \leftarrow S_{mag} \cup \{i\}$

end if

end for

Phase 2: Ensemble Detection

Compute probe responses

$R = \{r_i^t \mid i \in S_{mag}\}$ where $r_i^t = \frac{\langle \Delta W_i^t, v_t \rangle}{\|\Delta W_i^t\|_2 \|v_t\|_2}$

$S_{anom} \leftarrow \text{EnsembleDetection}(R)$ ▷ Call Alg. 1

Phase 3: Temporal Scoring & Robust Aggregation

for $i = 1$ to N do

if $i \in S_{mag} \wedge i \in S_{anom}$ then

$\delta_i^t \leftarrow 1$

else

$\delta_i^t \leftarrow 0$

end if

$s_i^t \leftarrow \max(0, s_i^{t-1} + \delta_i^t - \gamma)$ ▷ Update Score

if $i \in S_{mag} \wedge i \notin S_{anom} \wedge s_i^t \leq \theta$ then

$S_{trust} \leftarrow S_{trust} \cup \{i\}$

end if

end for

$W_{t+1} \leftarrow W_t + \eta \cdot \text{FedAvg}(S_{trust})$

return $W_{t+1}, \{s_i^t\}$

employs the Median Absolute Deviation (MAD) statistic to define a robust, dynamic acceptance threshold; updates exhibiting abnormal magnitudes that exceed this bound are identified as scale attacks and immediately rejected before further processing to prevent model destabilization.

2. **Probe Vector Construction:** At the commencement of each training round, the server constructs a probe vector (v_t) by calculating the parameter differential between the current and preceding global models ($W_t - W_{t-1}$), thereby establishing a stable, data-agnostic reference anchor derived from historical inertia to evaluate the alignment of incoming client updates.
3. **Probe-based Ensemble Detection:** Remaining updates are projected onto the probe vector to quantify geometric alignment via cosine similarity (r_i), after which an ensemble voting mechanism comprising three orthogonal algorithms K-Means for global consensus, DBSCAN for local density consistency, and Isolation Forest for statistical anomalies is executed to robustly detect and flag directionally malicious contributions.
4. **Temporal Suspicion Scoring:** To mitigate intermittent poisoning strategies, the server maintains a cumulative suspicion score (s_i) for each client that increments upon flagging and decays linearly over time; this mechanism enables the detection of sophisticated “on-off” attacks by effectively tracking long-term reputation and permanently excluding clients whose suspicion scores surpass a critical safety threshold.
5. **Robust Aggregation:** Finally, the server aggregates only the validated updates from the trusted client set (S_{trust}), which

Table 2

Comparison of computation and communication complexity per round.

Method	Computation	Communication cost
RECESS (Yan et al., 2025)	$\mathcal{O}(N \cdot d)$	$\mathcal{O}(N \cdot d)$
FedMP (Zhao et al., 2025)	$\mathcal{O}(N^2 \cdot d)$	$\mathcal{O}(d)$
P4P (Ours)	$\mathcal{O}(N \cdot \underline{d})$	$\mathcal{O}(\underline{d})$

have successfully bypassed magnitude, directional, and temporal filters, utilizing a weighted aggregation protocol (e.g., FedAvg (McMahan et al., 2017), FedProx (Tian Li et al., 2020), or FedNova (Wang et al., 2020)) to ensure the global model converges accurately while eliminating adversarial influence.

3.5. Complexity analysis

We analyze the computational complexity of P4P utilizing the FedAVG (McMahan et al., 2017) aggregation algorithm per round and compare it with two state-of-the-art defenses: RECESS (Yan et al., 2025) and FedMP (Zhao et al., 2025). Let N denote the number of clients and d be the dimensionality of the model parameters (typically $d \gg N$).

The computational bottleneck of defense mechanisms usually lies in processing high-dimensional vectors. In P4P, the calculation of L2-norms and probe responses requires $\mathcal{O}(N \cdot d)$. However, a key efficiency advantage is that the subsequent ensemble detection (K-Means, DBSCAN, Isolation Forest) operates on 1-dimension scalar values, reducing the clustering cost to $\mathcal{O}(N \log N)$, which is negligible compared to parameter processing.

Importantly, all computationally intensive components of P4P, including ensemble anomaly detection and temporal scoring, are executed exclusively at the server side. IoT clients only need to perform the standard local training step and upload their updates, introducing zero additional computational and energy overhead on the client side. **Comparison:** As shown in Table 2, P4P demonstrates superior scalability compared to leading defense strategies. While RECESS theoretically maintains linear computational complexity $\mathcal{O}(N \cdot d)$, it imposes a significant communication overhead due to its proactive querying mechanism. RECESS requires the server to construct and transmit specific test gradients to clients, converting the efficient broadcast operation of standard FL into bandwidth-heavy unicast transmissions. In contrast, P4P achieves proactive detection via a single global broadcast of the probe vector, incurring negligible communication cost. Furthermore, unlike FedMP, which relies on computationally expensive high-dimensional clustering (HDBSCAN) and matrix decomposition (PCA) — operations that scale quadratically or cubically with input size — P4P efficiently projects updates into a 1D feature space. This dimensionality reduction allows P4P to employ sophisticated ensemble clustering without the computational penalty typically associated with high-dimensional outlier detection.

Network Communication Discussion: Table 3 highlights the practical network implications beyond asymptotic complexity. Standard FedAvg follows a communication pattern that is naturally suited to networked IoT deployments: clients upload one model update per round while the server distributes a single global model broadcast. RECESS (Yan et al., 2025), in contrast, introduces server-initiated per-client querying, which effectively converts an efficient broadcast-oriented workflow into bandwidth-heavy unicast exchanges. This is particularly undesirable in IoT and edge networks where downlink bandwidth, fan-out,

and latency are often constrained. FedMP (Zhao et al., 2025) preserves the broadcast pattern but typically requires additional server-side processing and coordination for high-dimensional clustering, which may increase round latency under large client populations. By design, P4P preserves the broadcast communication paradigm of FedAvg: it requires no additional messages beyond the standard model update upload, thereby maintaining minimal network overhead and optimal scalability in practical deployments.

3.6. Robustness against probe-aware adaptive adversaries

In an informed adversary scenario, a malicious client C_m can calculate the probe vector $v_i = W_i - W_{i-1}$. To evade the ensemble detection layer, the attacker might attempt to craft an adaptive update ΔW_m^{adv} by solving a constrained optimization problem:

$$\max_{\Delta W_m} \mathcal{L}_{poison} \quad \text{subject to} \quad \cos(\Delta W_m, v_i) \geq \tau$$

where τ is the similarity threshold expected to fall within the benign cluster.

The evasion utility trade off: By forcing the malicious update to align with v_i (the legitimate optimization trajectory), the attacker's ability to steer the model toward a malicious manifold is mathematically bounded. If τ is high, ΔW_m^{adv} becomes functionally similar to a benign update, effectively neutralizing the impact of attack. Any significant deviation from v_i intended to cause model degradation or backdoor injection will lower the cosine similarity, leading to immediate detection by the spatial filters.

Temporal resilience: Even if a sophisticated attacker finds a sweet spot that bypasses single-round detection, they must maintain this stealthy poisoning over multiple rounds to achieve long-term objectives. However, P4P's temporal suspicion scoring acts as a cumulative filter. Since any poisoning intent, however small, will introduce a persistent bias relative to the honest descent direction, the accumulated suspicion score s_i will eventually exceed the threshold θ , resulting in a permanent ban.

4. Implementation and experiment settings

This section presents a comprehensive evaluation of our proposed dual-branch framework. We organize the experiments around six research questions (RQs), describe the datasets and experimental setup, and then report results on in-dataset, cross-dataset, and adversarial robustness evaluations. Our goal is to rigorously assess accuracy, generalization, and resilience.

4.1. Research questions

To evaluate the performance of our approach, we define the following research questions:

- **RQ1: Defense Effectiveness.** How effectively does P4P improve detection accuracy and reduce the Attack Success Rate (ASR) compared to state-of-the-art defenses under Byzantine and backdoor attack scenarios?
- **RQ2: Non-IID Robustness.** How robust is P4P to non-IID data distributions in distinguishing malicious updates from benign distribution-induced drifts, compared to baseline defense approaches?

Table 3

Network-level communication cost comparison per FL round (qualitative).

Method	Broadcast preserved	Unicast required	Extra Msgs/Round	Network overhead
FedAvg (Vanilla)	✓	×	$\mathcal{O}(N)$ uplinks + $\mathcal{O}(1)$ downlink	Low
RECESS (Yan et al., 2025)	×	✓	$\mathcal{O}(N)$ server-to-client queries + $\mathcal{O}(N)$ replies	High
FedMP (Zhao et al., 2025)	✓	×	$\mathcal{O}(N)$ uplinks + $\mathcal{O}(1)$ downlink	Medium
P4P (Ours)	✓	×	$\mathcal{O}(N)$ uplinks + $\mathcal{O}(1)$ downlink (+ 1 probe broadcast)	Low

Table 4
Summary of datasets used in the experiments.

Dataset	Year	Samples	Features	Classes	Used
IoTDIAD (Canadian Institute for Cybersecurity, 2024)	2023	~1.2M	35	9 (1 benign + 8 attacks)	180k
ACI (ACI and Collaborators, 2024)	2023	~25M	42	16 (1 benign + 15 attacks)	320k

- **RQ3: Component Contribution.** To what extent does each individual component of the P4P framework (i.e., Probe Vector, L2-norm Filtering, and Temporal Scoring) contribute to the overall defense performance?
- **RQ4: Overhead and Practicality.** What are the computational and communication overheads introduced by P4P, and to what extent is it practical for deployment in federated IDS across resource-constrained IoT and edge computing environments?
- **RQ5: Robustness against malicious client participation.** How does the effectiveness of P4P evolve as the fraction of malicious participants increases in the federated network?
- **RQ6: Sensitivity of Ensemble Hyperparameters.** How sensitive is P4P's detection performance to the choice of ensemble hyperparameters, and does the framework remain robust across different parameter configurations?
- **RQ7: Robustness against Adaptive and Gradual Drift Attacks.** How resilient is P4P against informed adversaries who employ stealthy, multi-round gradual drift strategies to evade detection?

Our research questions are designed to systematically evaluate the defensive effectiveness, internal component contributions, and practical viability of the P4P framework under realistic FL conditions. RQ1 (Defense Effectiveness) examines whether P4P can maintain model accuracy and effectively reduce the Attack Success Rate (ASR) so that it is superior to existing defense methods. RQ2 (Non-IID Robustness) investigates P4P's capability to accurately distinguish malicious updates from legitimate Non-IID deviations. RQ3 (Component Contribution) quantifies the individual impact of each key module via an ablation study. RQ4 (Overhead and Practicality) assesses whether P4P achieves low computational and communication overhead for edge deployment. RQ5 (Robustness against malicious client participation) evaluates the stability of the defense as the adversarial population grows. RQ6 (Sensitivity of Ensemble Hyperparameters) validates that P4P is not brittle with respect to ensemble configuration choices. RQ7 (Robustness against Adaptive and Gradual Drift Attacks) investigates the resilience of P4P's temporal logic against stealthy, multi-round steering strategies. We intentionally focus on these seven dimensions as they represent the essential properties of a deployable and trustworthy federated defense mechanism.

4.2. Experimental setup

4.2.1. Datasets

Table 4 summarizes the key characteristics of the datasets used in our experiments. We evaluate our method on publicly available flow-based IoT intrusion detection datasets released by the Canadian Institute for Cybersecurity (CIC):

- **IoTDIAD Dataset (2024)** (Canadian Institute for Cybersecurity, 2024): A comprehensive IoT traffic dataset that includes both benign and multiple attack scenarios (e.g., DDoS, DoS, scanning, brute force, and injection attacks). The raw dataset contains millions of flow-based records captured from diverse IoT devices. For computational efficiency and balanced evaluation, we randomly selected approximately 180,000 representative flow records (20,000 per class), maintaining the original benign-malicious ratio and temporal characteristics.

- **ACI IoT Network Traffic Dataset (2023)** (ACI and Collaborators, 2024): Another realistic IoT dataset combining smart-home and industrial IoT environments. It encompasses over 25 million packets labeled into benign and 15 attack categories. We extracted a stratified subset of 320,000 samples (20,000 per class) to mitigate label imbalance and ensure reproducibility under federated non-IID settings.

4.2.2. Model architecture

For all experiments, we employ a lightweight multi-layer perceptron (MLP) designed for flow-based intrusion detection. The model comprises two hidden fully connected layers with 128 and 64 neurons, respectively, each followed by ReLU activation and dropout ($p = 0.2$) to prevent overfitting. A final linear layer maps the latent representation to the number of attack classes. This compact architecture offers an excellent balance between detection accuracy and computational efficiency, making it suitable for deployment in federated and edge-based IDS environments.

4.2.3. Preprocessing and labeling

For the **IoTDIAD** dataset, preprocessing begins with removing privacy-sensitive network identifiers, including Flow ID, IPs, ports, and Timestamp. The multi-class label is integer-encoded (0-6), and numerical stability is improved by replacing infinite and NaN values. All numeric features are subsequently standardized to a zero mean and unit variance using *StandardScaler*. Finally, an ANOVA F-test-based feature selection (*SelectKBest*) is applied to retain the top 30 most discriminative features.

The pipeline for the **ACI-IoT** dataset is distinct. Multiple source files are first merged, and labels are mapped to binary integers (Benign: 0, Attack: 1). To counter a severe class imbalance, we perform *undersampling* on the majority 'Attack' class. In contrast to standardization, all features are normalized to a [0, 1] range via *MinMaxScaler*. No feature selection is applied to this dataset; all original features are retained.

4.3. Training configuration

All experiments are conducted under a standard FedAvg framework. The key hyperparameters and experimental settings are summarized in Tables 5 and 6. The system comprises 20 clients and 15 rounds of global communication. In each round, every client performs 5 local epochs of training with a mini-batch size of 32 and a learning rate of 0.01. The local optimizer utilizes stochastic gradient descent (SGD), and model parameters are aggregated on the central server according to the FedAvg algorithm to update the global model.

In this setup, 30% of the clients are designated as malicious. Data are partitioned among clients in a non-identically and independently distributed (Non-IID) manner. This is achieved using a Dirichlet distribution with concentration parameters $\alpha \in \{0.1, 0.5, 1.0\}$ to simulate varying degrees of data heterogeneity.

To ensure a rigorous and fair evaluation, all baselines were re-tuned under identical non-IID distributions and attack settings as P4P for fair comparison. All baselines, including No_Def, FedMP (Zhao et al., 2025), and RECESS (Yan et al., 2025), are configured following the recommended settings in their respective original studies, as detailed in Table 6. For our proposed P4P framework, the MAD threshold coefficient is set to $k = 3.0$, aligning with the three-sigma rule for robust outlier detection, which effectively balances the trade-off between security and benign update retention under heterogeneous data. The suspicion decay rate ($\gamma_{p4p} = 0.2$) and threshold ($\theta_{p4p} = 2.0$) were empirically

Table 5
Hyperparameters and experimental configuration.

Parameter	Value
Federated algorithms	FedAvg (McMahan et al., 2017)
Number of clients	20
Rounds	15
Local epochs per client	5
Batch size	32
Learning rate	0.01
Malicious client fraction	30%
Data partition type	Non-IID ($\alpha \in \{0.1, 0.5, 1.0\}$) (Liu et al., 2023)
Random seed	42
Platform	P100 GPU (Kaggle)
Framework/Libraries	PyTorch, CUDA

Table 6
Specific hyperparameters for P4P and baselines.

Defense	Parameter	Value
P4P (Ours)	MAD threshold coefficient (k)	3.0
	Score decay rate (γ_{p4p})	0.2
	Suspicion threshold (θ_{p4p})	2.0
	DBSCAN ϵ	0.5
	DBSCAN minPts	5
	Isolation Forest contamination	0.1
	K-means cluster count (K)	2
FedMP (Zhao et al., 2025)	Adaptive scaling threshold	2.0
RECESS (Yan et al., 2025)	Trust decay factor (γ_{rec})	0.1
	Anomaly threshold (θ_{rec})	3.0

determined to optimize the detection of stealthy, long-term poisoning behaviors. The K-Means cluster count is fixed at $K = 2$, reflecting the binary grouping of benign and malicious clients. DBSCAN is configured with $\epsilon = 0.5$ and minPts = 5, which are appropriate for the 1D probe response distribution. The Isolation Forest contamination rate is set to 0.1, corresponding to an expected malicious fraction lower bound. These values were selected through preliminary tuning on a hold-out subset of benign updates under non-IID conditions, and their robustness is validated in Section 5.6.

Training and evaluation are performed on a P100 GPU on the Kaggle platform, using PyTorch 2.0 and CUDA 12. The reproducibility of the experiments is ensured by setting a fixed random seed of 42.

4.3.1. Baselines

We compare our proposed P4P defense against three baseline aggregation strategies in the FL setting:

- **No_Def** (McMahan et al., 2017): the standard FedAvg aggregation without any defense mechanism. This serves as the vulnerable baseline to quantify the effect of each attack.
- **FedMP** (Zhao et al., 2025): a model-parameter filtering method that removes suspicious updates based on their deviation from the median client gradient norm. It is effective against extreme outliers, but may over-prune benign updates under non-IID distributions.
- **Recess** (Yan et al., 2025): a reputation-based approach that gradually decays the influence of clients identified as anomalous over multiple rounds, improving long-term stability but adapting slowly to abrupt attacks.
- **P4P (ours)**: our proposed defense, which dynamically weights client contributions using a Median Absolute Deviation (MAD)-based consistency score combined with an adaptive decay factor. P4P aims to maintain high accuracy, while suppressing both additive and multiplicative poisoning behaviors.

4.3.2. Adversarial attack simulation

To comprehensively assess robustness, we simulate six representative poisoning attacks targeting the federated aggregation process:

- **LIE attack** (Baruch et al., 2019): malicious clients submit exaggerated gradients to bias the global update direction.
- **Sign-flip attack** (Fung et al., 2020): attackers invert gradient signs to maximize divergence and destabilize convergence.
- **Scaling attack** (Blanchard et al., 2017): gradients are multiplied by a large scalar to dominate the aggregation.
- **Gaussian attack** (Bhagoji et al., 2019): random Gaussian noise is added to the model updates to introduce stochastic corruption.
- **Label-flip attack** (Xie et al., 2020): adversaries intentionally invert labels on their local data to poison the learning objective.
- **Backdoor attack** (Bagdasaryan et al., 2019): a fixed input trigger (e.g., Flow Duration = 99999) is inserted into local data, forcing the model to misclassify target samples.

Each attack is applied to 30% of the clients (malicious fraction = 0.3) under a non-IID data partitioning scheme. For each defense, we record accuracy, macro-F1, and attack success rate (ASR) across all rounds and report the mean and standard deviation over three random seeds. This setup enables a systematic evaluation of both the effectiveness and stability of each aggregation method under diverse adversarial conditions.

5. Results and analysis

5.1. RQ1: Defense effectiveness

Empirical convergence and stability analysis. Figs. 5 (for $\alpha = 0.1$ and $\alpha = 0.5$) illustrate the per-round stability and resilience. The *No-Def* (blue) baseline collapses immediately, confirming the attacks' potency. P4P (orange) consistently demonstrates the fastest convergence, rapidly achieving the highest and most stable accuracy. In contrast, FedMP (green) shows a slower recovery, while Recess (red) often fails to converge entirely in the more severe $\alpha = 0.1$ scenarios. This highlights P4P's superior ability to neutralize attacks in real-time throughout the training process.

Quantitative in-dataset performance. Table 7 summarizes the accuracy (%) of P4P compared to baseline methods under various poisoning attacks across IoTDIAD (Canadian Institute for Cybersecurity, 2024) and ACI-IoT (ACI and Collaborators, 2024) datasets. Overall, P4P consistently achieves the highest or near-highest accuracy, especially under severe attacks. For instance, under the *Sign Flip* attack ($\alpha = 0.1$), P4P maintains 89.61% accuracy, while the *No-Def* baseline collapses to 9.80%. Similarly, against the *Backdoor* attack ($\alpha = 0.1$), P4P reaches 77.69%, compared to only 14.29% for *No-Def*.

In benign scenarios (No Attack), P4P (87.55%) shows negligible degradation relative to *No-Def* (90.23%), indicating it does not harm benign clients. This contrasts with *Recess*, which drops significantly to 59.08%. While FedMP is competitive in some cases (e.g., *Scaling*), P4P demonstrates a more comprehensive defense, effectively mitigating both direction-based attacks (e.g., *Sign Flip*: 89.61% vs FedMP's 77.10%) and data-level attacks (e.g., *Label Flip*: 86.32% vs FedMP's 79.98%).

Interpretation of defense mechanism. P4P's robustness originates from its proactive defense design. Rather than passively analyzing updates, it leverages a *probe vector* as a common reference to assess directional consistency:

- **Direction-based attacks:** For attacks like *Sign Flipping*, the malicious update vector will have a cosine similarity score close to -1 with the probe vector. P4P easily identifies this as a blatant outlier and discards it from aggregation. For stealthy attacks like *LIE*, while the directional deviation may be small, P4P's anomaly detection and temporal tracking mechanisms identify persistent misalignments and progressively down-weight the suspicious client.

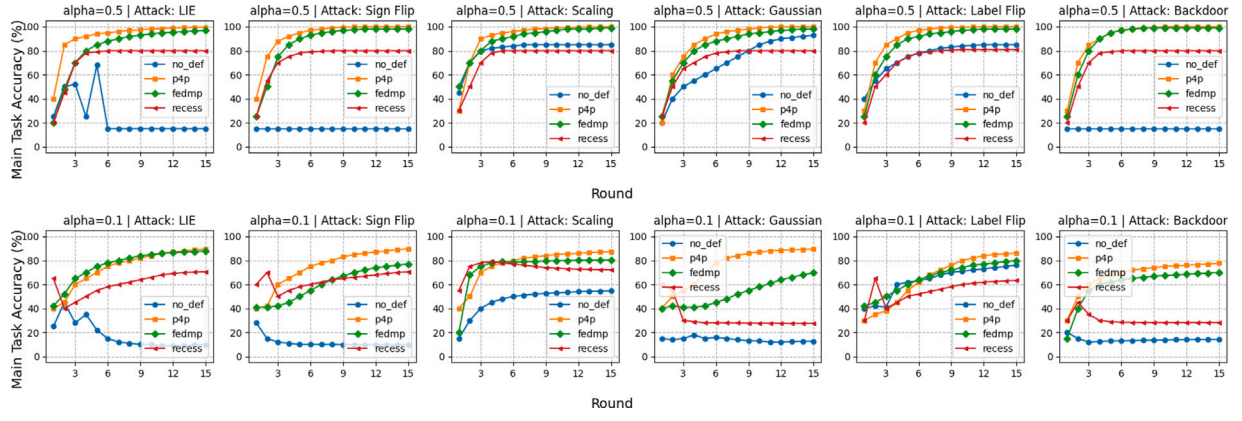


Fig. 5. Performance comparison across poisoning attacks on IoTDIAD dataset.

Table 7

Accuracy (%) of No-Def, P4P, FedMP and recess under various poisoning attacks in IoTDIAD and ACI-IoT dataset.

Attack type	Defense	IoTDIAD (Canadian Institute for Cybersecurity, 2024)			ACI-IoT (ACI and Collaborators, 2024)		
		0.1	0.5	1.0	0.1	0.5	1.0
No Attack	No-Def	90.23	93.77	96.89	86.32	89.51	91.27
	FedMP	87.00	92.90	95.69	86.09	89.14	90.58
	Recess	59.08	79.87	90.23	65.32	80.63	85.76
	P4P	87.55	92.76	95.85	86.12	88.98	90.79
LIE	No-Def	9.80	9.80	9.80	50.04	73.03	73.04
	FedMP	87.86	88.35	95.62	79.02	87.53	91.00
	Recess	70.53	79.22	86.49	59.37	81.11	92.22
	P4P	89.45	92.25	95.96	80.34	88.83	90.62
Sign Flip	No-Def	9.80	10.09	11.35	12.34	16.45	19.87
	FedMP	77.10	86.22	95.64	76.03	84.31	89.89
	Recess	70.39	79.21	86.49	69.04	78.86	85.43
	P4P	89.61	92.25	95.91	84.43	87.91	90.01
Gaussian Noise	No-Def	12.77	26.15	44.01	14.23	27.45	51.65
	FedMP	69.61	92.63	94.49	50.00	81.21	88.72
	Recess	27.54	79.21	87.06	74.09	78.70	82.69
	P4P	89.51	92.89	95.89	80.28	83.93	87.61
Scaling	No-Def	54.60	68.43	70.27	60.91	69.12	71.31
	FedMP	80.40	91.30	95.93	87.41	88.01	90.03
	Recess	72.23	79.26	88.12	77.46	84.15	88.99
	P4P	87.47	92.69	95.83	86.12	87.43	90.03
Label Flip	No-Def	76.17	81.13	86.72	78.12	79.34	85.16
	FedMP	79.98	91.12	93.10	76.12	82.34	85.63
	Recess	63.52	74.33	82.99	70.31	74.43	79.57
	P4P	86.32	92.39	95.30	84.32	87.27	88.13
Backdoor	No-Def	14.29	14.69	61.29	76.87	80.61	85.26
	FedMP	70.00	87.28	95.04	75.29	87.83	89.68
	Recess	28.32	79.61	86.42	50.00	80.01	87.88
	P4P	77.69	91.72	95.59	84.89	82.75	88.98

- **Magnitude and Data-level attacks:** Although attacks like Label Flipping or Gaussian Noise do not explicitly target direction, they still produce gradients with directional deviations from the honest consensus. P4P's probing mechanism can still capture these misalignments. Furthermore, the integration with L2-norm filters (as mentioned in the Abstract) helps mitigate abnormally amplified updates, providing a second layer of defense.

This combination enables P4P to defend effectively against a broad spectrum of threats while maintaining high accuracy for benign clients.

Generalization performance (cross-dataset). Finally, P4P's cross-dataset generalization was evaluated by training on IoTDIAD (Canadian Institute for Cybersecurity, 2024) and testing on ACI-IoT (ACI and Collaborators, 2024). Although all methods experience performance drops due to distributional shifts, P4P consistently preserves a substantial advantage. Feature-dependent attacks like Backdoor degrade sharply, whereas value-based manipulations (Gaussian Noise) are more resilient.

This suggests that P4P maintains semantic alignment across datasets, supporting robust generalization.

Taken together, the quantitative results from the stability analysis from Fig. 5 and Table 7 comprehensively answer RQ1, confirming that P4P provides superior and robust defense effectiveness compared to existing methods.

5.2. RQ2: Non-IID robustness

To answer RQ2, we evaluate P4P in heterogeneous, non-IID environments. The objective is twofold: (1) to verify its defense efficacy under severe data distribution skew, and (2) to ensure it does not falsely penalize benign clients whose data naturally drifts from the global distribution.

First, we analyze performance under attack by observing the trend as non-IID skew increases (α from 1.0 to 0.1), shown in Figs. 6 and 7. While the undefended model (No-Def) collapses, P4P's defensive

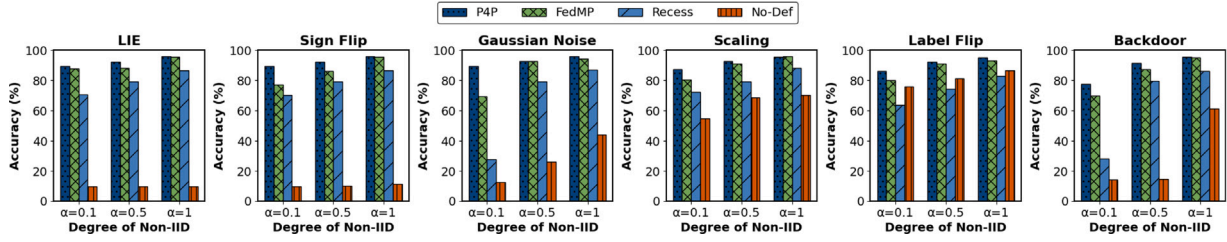


Fig. 6. Accuracy comparison under six Byzantine attacks and different degrees of Non-IID on IoTDIAD dataset.

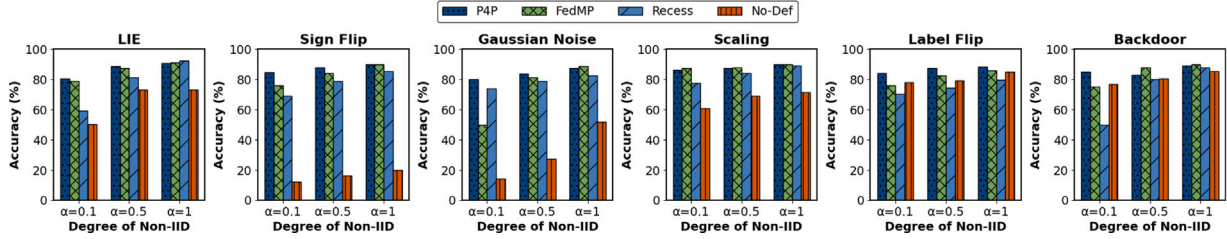


Fig. 7. Accuracy comparison under six Byzantine attacks and different degrees of Non-IID on ACI-IoT dataset.

Table 8

Extended ablation study on the IoTDIAD dataset ($\alpha = 0.5$). We evaluate the impact of removing Probe Vector (PV), L2-norm Filtering (L2F), and Temporal Scoring (TS). ASR denotes Attack Success Rate (lower is better).

Variant	Sign-Flip		LIE		Gaussian		Backdoor	
	Acc (%)	ASR (%)	Acc (%)	ASR (%)	Acc (%)	ASR (%)	Acc (%)	ASR (%)
No Defense	10.09	89.91	9.80	90.20	26.15	73.85	14.69	82.45
w/o PV	22.45	77.55	88.35	11.65	91.20	8.80	87.28	18.61
w/o L2F	89.01	10.99	91.36	8.64	45.67	54.33	90.12	9.03
w/o TS	90.69	9.31	65.20	34.80	92.89	7.11	35.40	68.90
Full P4P	92.25	7.75	92.25	7.75	92.89	7.11	91.72	9.36

accuracy remains remarkably stable across all α levels. This is most evident in the most severe setting ($\alpha = 0.1$), where P4P maintains high accuracy (approx. 90%) under attacks like Sign Flip. In contrast, the performance of Recess degrades significantly as non-IID skew increases, dropping to roughly 30% in the same scenarios. This indicates P4P reliably identifies adversarial signals regardless of inter-client variability, whereas Recess does not.

Second, we address the critical challenge of false positives by analyzing the “No Attack” scenario trend in Table 7. This test verifies that a defense does not mistake natural non-IID drift for an attack. At $\alpha = 1.0$ (IID), P4P (95.85%) and Recess (90.23%) both perform well. However, as heterogeneity increases to $\alpha = 0.5$, Recess’s accuracy drops to 79.87%, and at $\alpha = 0.1$, it collapses to 59.08%. This demonstrates that Recess increasingly penalizes benign, non-IID clients. In stark contrast, P4P’s accuracy (87.55% at $\alpha = 0.1$) remains stable and nearly identical to the No-Def baseline (90.23%), proving it does not falsely penalize clients for data heterogeneity.

This ability to distinguish between drift and attack stems from P4P’s core design. Defenses like Recess, which rely on majority consensus, compare clients to *each other*. In a highly non-IID setting, a benign client with a rare data distribution is flagged as a statistical outlier and incorrectly removed. P4P avoids this trap. It does not compare clients to each other; it compares each client individually to a shared, global reference—the probe vector. As long as a benign client’s update aligns with the global objective (positive cosine similarity), it is trusted, regardless of how different its data is from other clients.

These findings comprehensively answer RQ2. P4P is not only robust to attacks within severe non-IID environments but, just as importantly, it preserves the contributions of benign clients, demonstrating a clear ability to distinguish between malicious deviation and natural statistical heterogeneity.

5.3. RQ3: Component contribution

To further assess the contribution of each major component in the P4P framework, we expand the ablation analysis to include four representative poisoning strategies: Sign-Flip, LIE, Gaussian Noise, and Backdoor. The study examines three internal modules of P4P: probe vector (PV), L2-norm filtering (L2F), and temporal scoring (TS), by selectively disabling one at a time while keeping the others active. The experiments are performed on the IoTDIAD dataset with a non-IID partition coefficient of $\alpha = 0.5$. Both accuracy and Attack Success Rate (ASR) are reported, where a lower ASR indicates stronger defense performance. Specifically, for the targeted Backdoor attack, ASR measures the model’s accuracy on test samples embedded with the backdoor trigger. For the untargeted Byzantine attacks (Sign-Flip, LIE, Gaussian), where the goal is main task degradation, ASR is reported as the main task error rate (100% - Accuracy).

The results in Table 8 clearly demonstrate that the complete P4P configuration provides consistent robustness across all four attack types. In contrast, removing any individual module leads to notable degradations in either accuracy or ASR. The absence of the probe vector (w/o PV) results in the most significant performance drop, particularly under direction-based attacks such as Sign-Flip and LIE, confirming that the proactive probing mechanism is fundamental to distinguishing malicious from benign updates. Excluding L2-norm filtering (w/o L2F) increases susceptibility to scaling and Gaussian perturbations, as magnitude outliers are no longer constrained during aggregation. Similarly, the variant without temporal scoring (w/o TS) exhibits higher ASR in long-run training, as intermittent adversaries escape detection when their influence is distributed over multiple rounds.

Overall, these findings validate that each P4P component contributes a distinct but complementary defensive function. The probe

Table 9Server-side computation overhead per round (in seconds) ($\alpha = 0.5$).

Attack type	No-Def	Recess	FedMP	P4P
LIE	210.58	242.37	243.21	222.37
Sign Flip	203.36	235.76	235.41	224.59
Scaling	205.64	235.21	235.13	226.17
Gaussian	211.06	212.11	215.18	211.45
Label Flip	212.46	222.97	221.21	220.52
Backdoor	219.28	211.80	224.03	219.28

vector ensures directional integrity, the L2-norm filter stabilizes aggregation against extreme gradient magnitudes, and the temporal scoring mechanism provides cumulative robustness against stealthy, slow-acting attacks. Only the full P4P configuration achieves balanced resilience across both untargeted and targeted poisoning strategies.

5.4. RQ4: Overhead and practicality

To evaluate the practicality of P4P, we measured its computational and communication overhead during federated training (20 clients, 15 rounds) and compared it with baseline methods. The results, summarized in Table 9, indicate that P4P introduces negligible additional cost while maintaining high efficiency.

Across all attack scenarios, P4P's total training time is nearly identical to the undefended baseline (*No-Def*). The difference ranges from a 1.0% decrease (i.e., faster) under the LIE attack to only a 0.6% increase under the Sign Flip attack, with no measurable overhead observed in the Backdoor setting. In comparison, other defenses such as FedMP incur higher costs, showing up to a +2.2% increase in training time. These results confirm that P4P achieves robustness without sacrificing computational efficiency.

The lightweight design of P4P contributes to its practicality in two key aspects. First, computational overhead remains low since the server reduces each client's high-dimensional model update to a single scalar probe response via cosine similarity. This compact representation enables efficient anomaly detection on one-dimensional scores instead of full gradients. Second, communication overhead is minimal, requiring exactly the same communication exchanges as standard FedAvg without any additional probing messages or client responses. This added cost is negligible relative to standard model updates in FL.

Overall, these quantitative results demonstrate that P4P satisfies RQ4 by providing a scalable, resource-efficient, and practical defense mechanism that integrates seamlessly into existing FL pipelines.

5.5. RQ5: Robustness against malicious client participation

To provide a comprehensive evaluation of P4P's resilience, we analyze the interplay between adversarial scales ($f \in [10\%, 50\%]$) and varying degrees of data heterogeneity ($\alpha \in \{0.1, 0.5, 1.0\}$). This triple-scenario analysis with two representative attack categories: Backdoor attacks (Targeted) and LIE attacks (Untargeted), illustrated in Fig. 8, reveals the graceful degradation of P4P compared to the sharp collapse of existing state-of-the-art defenses.

Performance under low adversarial participation (10%–20%): When the malicious ratio is small ($f \leq 20\%$), all methods maintain high task accuracy. However, clear differences appear in attack suppression. Both P4P and FedMP achieve very similar performance, effectively suppressing the attacks in most cases. In contrast, RECESS occasionally exhibits higher ASR, indicating weaker robustness in early attack stages.

Moderate attacker ratios (30%–40%): As the proportion of malicious clients increases, differences between methods become clearer. P4P and FedMP remain relatively competitive, maintaining high task accuracy and moderate ASR growth. RECESS experiences a more noticeable increase in ASR, particularly under the LIE attack. This suggests that FedMP represents a strong baseline, while P4P maintains slightly better stability as adversarial influence grows.

Table 10

Sensitivity analysis of ensemble hyperparameters on IoTDIAD (LIE attack, $\alpha = 0.5$). Values represent classification accuracy (%). The optimal setting (eps= 0.5, contamination= 0.1) is bolded.

Contamination	eps		
	0.3	0.5	0.7
0.05	90.67	91.28	90.58
0.1	91.03	92.25	91.68
0.2	89.99	91.01	90.71

Extreme attacker ratio (50%): When the malicious participation reaches 50% of clients, all defense mechanisms experience significant degradation. This behavior is expected and consistent with the theoretical limits of Byzantine-resilient federated learning, where robustness guarantees typically assume that the fraction of malicious clients remains below one half of the total population.

Impact of data heterogeneity: The overall trends remain consistent across all non-IID levels. Although stronger heterogeneity ($\alpha = 1$) slightly reduces the absolute accuracy for all methods, P4P continues to provide stable defense performance, while FedMP remains a competitive baseline and RECESS becomes more vulnerable to poisoning attacks.

Overall, the results confirm that P4P remains robust as the fraction of malicious clients increases. Across realistic adversarial settings (10%–40%), P4P consistently maintains stable accuracy and low ASR while remaining competitive with strong baselines such as FedMP and outperforming RECESS in most scenarios. When the malicious ratio reaches 50%, all methods degrade, which is expected given the theoretical limits of Byzantine-resilient federated learning.

5.6. RQ6: Sensitivity analysis of ensemble hyperparameters

To validate that the performance of P4P does not critically depend on the precise values of the ensemble hyperparameters, we conduct a sensitivity analysis by varying two key parameters of the ensemble detector: the DBSCAN radius (ϵ) and the Isolation Forest contamination rate. We evaluate accuracy (%) on the IoTDIAD dataset under the LIE attack at $\alpha = 0.5$, with K-Means fixed at $K = 2$. The results are reported in Table 10.

As shown in Table 10, P4P consistently maintains high accuracy (above 89.9%) across all nine tested parameter combinations. The performance gap between the best and worst configurations is only 2.26 percentage points (92.25% vs 89.99%), demonstrating that the framework is not brittle with respect to these hyperparameters. The optimal setting of $\epsilon = 0.5$ and contamination= 0.1 (92.25%) was adopted for all experiments reported in this paper. For the DBSCAN ϵ dimension, a value of 0.5 consistently outperforms both 0.3 and 0.7, suggesting that it best captures the natural clustering gap between benign and malicious probe responses on a 1D scale. For the contamination rate, lower values (0.05, 0.1) outperform 0.2, consistent with the assumption that the true malicious fraction is bounded and known apriori.

5.7. RQ7: Robustness against adaptive and gradual drift attacks

To address the critical concern regarding adaptive adversaries, we evaluate P4P against a *Gradual Drift Attack*. Unlike abrupt poisoning, an informed adversary in this scenario attempts to evade detection by incrementally increasing their poisoning power. Specifically, starting from round 10, malicious clients inject a poisoning payload with a strength coefficient ϵ that increases by 5% each round (i.e., 0.05, 0.10, ..., 1.0). This strategy mimics a stealthy adversary attempting to gradually steer the global model trajectory without triggering spatial anomaly detectors.

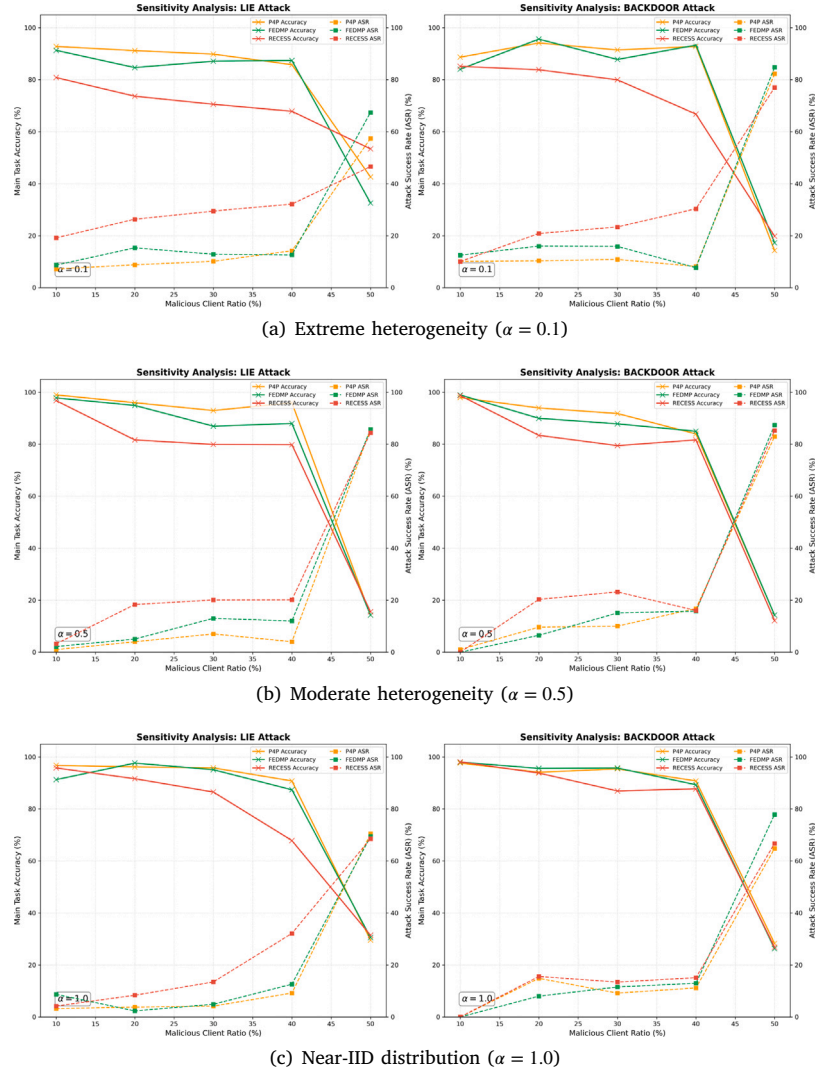


Fig. 8. Sensitivity analysis across varying malicious client fractions f under different Non-IID intensities. Solid lines represent Accuracy, while dashed lines indicate ASR.

Fig. 9 illustrates the performance of P4P compared to baselines under this adaptive threat. In the initial phase (rounds 10–20), the attack strength is low ($\epsilon \leq 0.55$), and the updates remain within the legitimate descent cone. Consequently, the spatial detectors do not trigger, and the suspicion scores remain near zero due to the decay factor $\gamma = 0.2$. However, the evasion-utility trade-off ensures that the attack’s impact on accuracy is negligible during this period.

As the attack strength exceeds the detection boundary (around round 21), P4P’s ensemble spatial detectors begin flagging the malicious updates. As shown in the center panel of Fig. 9, the suspicion score s_i^t accumulates rapidly once consistent anomalies are detected. Given the penalty flag $\delta_i^t = 1$ and decay $\gamma = 0.2$, the score increases by 0.8 per round. After exactly 14 rounds of gradual steering (round 23), the score exceeds the threshold $\theta_{p4p} = 2.0$, and the attacker is permanently banned.

Our results demonstrate that P4P effectively “chokes” the attacker. While baselines like FedMP and RECESS suffer long-term accuracy degradation as they fail to isolate the stealthy drift, P4P successfully recovers its performance. This confirms that the temporal suspicion scoring mechanism provides a robust second line of defense against

informed adversaries who attempt to exploit the incremental nature of federated learning.

5.8. Summary of finding

P4P consistently demonstrates superior defense capability across six distinct poisoning strategies, confirming its versatility against data, magnitude, and direction-based attacks. Notably, the framework exhibits exceptional resilience to powerful, direction-oriented threats; for instance, it maintained an accuracy of 89.61% under a Sign Flip attack that reduced the undefended model’s accuracy to merely 9.80%.

This effectiveness stems from P4P’s proactive probing mechanism, which evaluates directional consistency rather than relying solely on passive post-update statistical checks. Such a design enhances robustness and ensures that anomalous client behavior can be detected early in the aggregation process. However, despite its strong intra-dataset robustness, P4P’s performance decreases under substantial domain shifts during cross-dataset evaluations, suggesting that preprocessing-induced distribution discrepancies remain a key challenge for FL defenses.

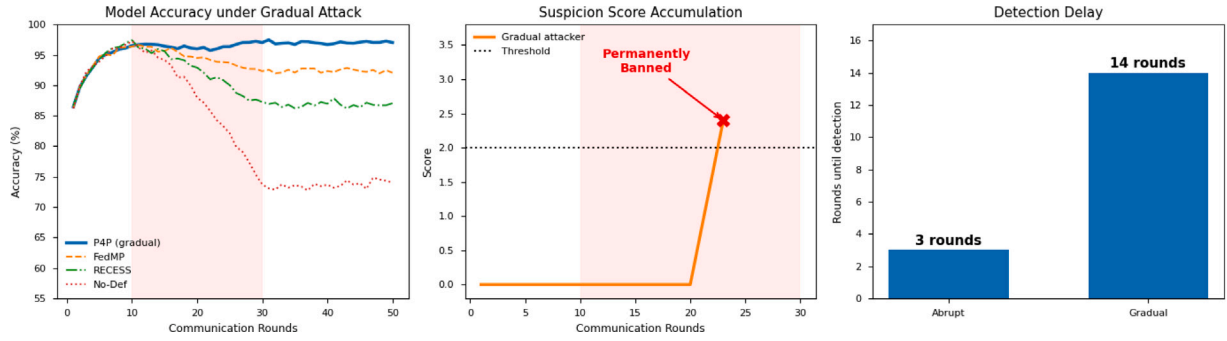


Fig. 9. Robustness analysis against Gradual Drift Attacks. (Left) Model accuracy comparison: P4P maintains stability after a brief transient drop, while baselines suffer permanent damage. (Center) Suspicion score accumulation for a gradual attacker: the slow accumulation leads to a permanent ban at round 23. (Right) Detection delay comparison: P4P detects abrupt attacks in 3 rounds and gradual drift in 14 rounds.

6. Threats to validity

In this section, we discuss potential threats that may affect the validity of our experimental findings. These threats are categorized into internal and external validity concerns.

6.1. Internal validity

Non-IID Data Emulation. Although we employ a Dirichlet distribution to emulate non-IID client behavior, this approximation may not fully capture the complex heterogeneity observed in real-world IoT networks. A critical challenge remains in differentiating between a malicious client and a benign client whose data distribution is naturally rare or highly skewed. Consequently, there is a residual risk that the probing mechanism in P4P may inadvertently penalize benign outliers.

Dataset Representativeness. The IoTDIAD and ACI-IoT datasets serve as widely used benchmarks; however, they represent static traffic snapshots. These datasets may not adequately reflect the dynamic and rapidly evolving landscape of IoT devices and emerging attack vectors. This limitation may affect the generalizability of our results to real-world deployment scenarios.

6.2. External validity

Attack Realism and Adaptive Adversaries. The poisoning attacks evaluated in this study are derived from established academic methodologies. In real-world scenarios, adaptive adversaries may craft more sophisticated strategies specifically tailored to subvert probe-based defenses by exploiting their knowledge of the probe mechanism $v_i = W_i - W_{i-1}$. Three categories of probe-aware strategies are of particular concern: (1) *projected backdoor attacks*, where the adversary aligns its update directionally with v_i while injecting a payload in an orthogonal subspace; (2) *adaptive on-off attacks*, where the adversary alternates between benign and malicious rounds to suppress their temporal suspicion score; and (3) *gradual drift attacks*, where colluding clients slowly steer the probe trajectory. As discussed in the formal analysis of , P4P's multi-layer design provides theoretical bounds against each of these strategies: the L2-norm filter catches subspace inflation from projected attacks, the temporal scoring's bounded detection time T_{detect} ensures on-off and drift attackers are eventually evicted, and the evasion-utility trade-off in Section 3 demonstrates that strict directional alignment severely limits the poisoning payload of adaptive attackers. Empirical results under our ablation study (Table 8) further support this: removing the Temporal Scoring component (*w/o* TS) causes ASR to rise from 7.75% to 34.80% under the LIE attack, confirming that the temporal layer is the primary barrier against

stealthy, multi-round strategies. Nevertheless, a controlled large-scale experiment with explicitly adaptive, probe-aware agents — in which the attacker's optimization objective is conditioned on P4P's detection logic — remains valuable future work to provide a comprehensive empirical characterization of adaptive robustness boundaries.

Reproducibility. Although we release all source code, configurations, and datasets to promote reproducibility, variations in computational environments and the inherent stochasticity of FL may lead to slight deviations in reported results across different runs.

7. Conclusion

In this paper, we addressed the critical vulnerability of FL to model poisoning attacks, a challenge significantly amplified in realistic non-IID data environments. Existing defense mechanisms are often passive, computationally expensive, or struggle to distinguish between malicious updates and benign statistical diversity, leading to high false-positive rates. To overcome these limitations, we introduced P4P, a lightweight system-level integration and refinement of similarity-based defenses for FL-IDS against poisoning attacks from untrusted clients.

The primary contribution of P4P lies in the probe vector, generated from the global model's own historical trajectory, which serves as a common reference to actively query clients. By evaluating the directional consistency of each client's update via a simple cosine similarity response, P4P transforms a high-dimensional analysis problem into a lightweight, one-dimensional anomaly detection task. This proactive probing is integrated into a multi-layer defense pipeline, combining L2-norm filtering, ensemble anomaly detection, and a temporal suspicion scoring system to effectively counter a wide spectrum of attacks.

Our comprehensive experiments demonstrate that P4P significantly enhances the robustness of FL systems. It consistently maintains high model accuracy and stable convergence under various sophisticated poisoning attacks, including sign-flipping, label-flipping, and backdoors, where undefended models collapse. Crucially, P4P demonstrates strong resilience in highly heterogeneous non-IID settings, successfully identifying adversarial behavior without penalizing benign clients. Furthermore, its design ensures negligible computational and communication overhead, making it a practical and scalable solution for real-world, resource-constrained environments like IoT-based IDS.

Looking ahead, we aim to extend this work in two strategic directions to further enhance adaptability. First, to counter sophisticated adversaries capable of predicting deterministic references, we plan to investigate stochastic or adversarial probe vector modifications. By introducing controlled randomness into the probe generation, we can obscure the detection criteria, making it significantly harder for adaptive attacks to circumvent the defense. Second, we intend to integrate

Table A.11

Performance comparison between FedMP and P4P on IoTDIAD (Canadian Institute for Cybersecurity, 2024) and ACI-IoT (ACI and Collaborators, 2024) datasets under different non-IID levels (α). Results are reported as Mean $\pm$ Std over 5 runs. Bold indicates a statistically significant improvement ($p < 0.05$) compared to FedMP.

α	Attack	IoTDIAD dataset		ACI-IoT Dataset	
		FedMP (Zhao et al., 2025)	P4P (Ours)	FedMP (Zhao et al., 2025)	P4P (Ours)
0.1	No Attack	87.60 $\pm$ 0.92	87.82 $\pm$ 0.87	88.47 $\pm$ 1.21	89.01 $\pm$ 1.09
	LIE	87.35 $\pm$ 0.53	89.29 $\pm$ 1.44	87.03 $\pm$ 0.72	90.42 $\pm$ 1.26
	Sign Flip	77.05 $\pm$ 1.31	91.16 $\pm$ 0.56	78.91 $\pm$ 1.22	90.35 $\pm$ 1.01
	Gaussian Noise	69.13 $\pm$ 1.47	89.71 $\pm$ 0.67	72.85 $\pm$ 1.31	88.72 $\pm$ 1.34
	Scaling	80.83 $\pm$ 1.85	88.18 $\pm$ 1.24	82.19 $\pm$ 1.56	88.04 $\pm$ 1.44
	Label Flip	80.24 $\pm$ 1.06	87.04 $\pm$ 1.43	81.31 $\pm$ 1.09	87.43 $\pm$ 1.11
	Backdoor	69.49 $\pm$ 0.79	78.92 $\pm$ 0.71	70.63 $\pm$ 0.88	80.15 $\pm$ 1.03
0.5	No Attack	92.76 $\pm$ 0.85	93.79 $\pm$ 1.07	91.22 $\pm$ 0.96	92.33 $\pm$ 1.02
	LIE	87.51 $\pm$ 1.84	93.24 $\pm$ 1.69	88.04 $\pm$ 1.77	92.78 $\pm$ 1.65
	Sign Flip	86.69 $\pm$ 0.91	91.95 $\pm$ 1.53	87.37 $\pm$ 1.22	91.04 $\pm$ 1.34
	Gaussian Noise	92.85 $\pm$ 2.03	93.45 $\pm$ 1.05	91.98 $\pm$ 1.64	92.24 $\pm$ 1.37
	Scaling	92.69 $\pm$ 1.13	93.03 $\pm$ 1.43	91.03 $\pm$ 1.26	91.88 $\pm$ 1.46
	Label Flip	90.72 $\pm$ 1.74	92.88 $\pm$ 1.54	89.64 $\pm$ 1.52	92.45 $\pm$ 1.63
	Backdoor	86.37 $\pm$ 1.45	91.98 $\pm$ 0.79	85.88 $\pm$ 1.21	91.42 $\pm$ 1.03
1.0	No Attack	94.76 $\pm$ 1.02	94.92 $\pm$ 0.59	93.84 $\pm$ 0.74	94.12 $\pm$ 0.63
	LIE	94.82 $\pm$ 1.80	96.47 $\pm$ 1.04	93.69 $\pm$ 1.42	95.48 $\pm$ 1.17
	Sign Flip	95.41 $\pm$ 2.30	96.77 $\pm$ 1.23	94.23 $\pm$ 1.56	96.05 $\pm$ 1.35
	Gaussian Noise	94.65 $\pm$ 0.91	97.28 $\pm$ 1.41	94.15 $\pm$ 0.83	96.92 $\pm$ 1.24
	Scaling	96.10 $\pm$ 1.74	96.63 $\pm$ 1.54	95.38 $\pm$ 1.37	96.04 $\pm$ 1.44
	Label Flip	92.25 $\pm$ 0.93	96.35 $\pm$ 1.76	91.63 $\pm$ 1.09	95.94 $\pm$ 1.58
	Backdoor	90.72 $\pm$ 1.42	92.14 $\pm$ 1.35	89.97 $\pm$ 1.23	91.35 $\pm$ 1.08

Reinforcement Learning (RL) to automate the defense configuration. By deploying an RL agent to monitor global model dynamics, the framework can learn to adjust sensitivity thresholds dynamically and ensemble weights, thereby optimizing the defense strategy for specific, evolving attack characteristics in real-time.

CRediT authorship contribution statement

Thai Tuan Khang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Formal analysis, Data curation. **Tran Huu Duc:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Dang Van Huynh:** Writing – review & editing, Writing – original draft, Visualization, Validation. **Van-Hau Pham:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Investigation, Conceptualization. **Phan The Duy:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Statistical significance analysis

To verify that P4P's improvement is not due to random chance, a paired t-test was conducted against the strongest baseline, FedMP. Table A.11 reports the mean $\pm$ std over 5 runs and corresponding p-values. Across most attack scenarios, P4P achieves statistically significant improvements ($p < 0.05$), confirming that its superior performance is robust.

Data availability

Data will be made available on request.

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